# Physics of Buddhism

## The physics and mathematics foundation of Buddhism

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**Abstract**

Religion and Science have long been antagonistic to one another in which the former is not experimentable; while the latter requires precise logical reasoning, theoretical construction and experimental verification. Although in recent decades, dialogues between the two subjects have been performed, a formal scientific theory of religion has never been established. In this paper, we present the physics and mathematics of Buddhism for the first time in history and show that the theory in Buddhism can be elegantly expressed in physics and mathematics, in particular quantum mechanics and quantum field theory. Several core foundations in Mahayana Buddhism, including the thoughts of Mādhyamaka and Yogācāra School of Buddhism are compatible with physics.

This paper inherits the work of our previous paper-Theory of Duality in reference [1]. The mathematical formalism of duality, which infers as the mathematical foundation of Taoism, serves as the building ground of the physics theory of Buddhism. The parity group $\mathbb{Z}_2$ and the Klein-4 group $\mathbb{Z}_2 \times \mathbb{Z}_2$ and their representations are fundamental in developing the mathematical theory of Taoism and Buddhism. We give the physics and mathematical definition of thought as a duality wave, upon second quantization it will produce quanta of thought, which shows matter appearance in the matter domain. Finally the partition function and the average energy of the duality wave system are computed.

# Contents

# Chapter 1

# Introduction

The development of science and physics has been remarkably successful in the last 300 years. Numerous successful theories have been developed by notable physicists, from the classical world of Newtonian celestial mechanics, to quantum mechanics founded by Schrödinger and Heisenberg and some others which describes the microscopic atomic world. Later, Dirac and Feynman promoted quantum mechanics to quantum field theory [2, 3, 5, 4, 6, 7], which lays the paradigm of the standard model of sub-atomic particles. Later, C.N.Yang and R.Mills developed the celebrated Yang-Mills theory which provides the mathematical foundation for non-abelian gauge field interaction [8], which facilitates the discovery of quantum chromodynamics by Gell-Mann [9] and electroweak unification by Weinberg [10]. Finally, the discovery of Higgs boson explains the mass of electroweak gauge bosons and the acquisition of mass of leptons and quarks [11, 12, 13, 14, 15]. And attributed to Einstein, the theory of general relativity is the complete theory of gravity up to date which describes the macroscopic world and has a lot of implications to modern cosmology.

Despite the success of physics and science, there are still a lot of mysteries that are unanswered and yet to be discovered. Dark matter contributes to 25% of the mass but its nature is still unknown [16, 17]. The question on the fine tuning problem in particle physics, the Hierarchy problem of the Higgs Boson remains a serious issue,

[redacted]

Figure 1.1: The Hierarchy problem addresses the fine tuning problem of renormalized Higg's mass. The Higg's mass is corrected by higher-order loop processes which are ultraviolet (UV) divergent.

$$\tilde{\mathrm{m}}_{\mathrm{H}}^2 = m_{\mathrm{H}}^2 + \frac{\Lambda^2}{32\pi^2}(4\lambda^2 + 3g'^2 + 9g_W^2 - 24Y_{ij}^{f\,2}) + \mathcal{O}(\lambda^2, g'^4, g_W^4, Y_{ij}^{f\,4})\,, \tag{1.1}$$

where $\tilde{\mathrm{m}}_{\mathrm{H}}^2$ is the renormalized Higg's mass and the $m_{\mathrm{H}}^2$ is the bare Higg's mass, other parameters with $\lambda$ the Higgs self coupling, $g'$ is the coupling of the $Z$ boson, $g_W$ is the coupling of the $W$ boson and $Y_{ij}^f$ is the Yukawa coupling [18]. In addition, the origin of the 18 independent parameters in the standard model of particles remains the hardest and deepest problems in physics.

Worse still, physics and science do not address theories and laws in religion and humanities. Conciousness is also an issue that arouse serious debates, although some researches have been conducted by Sir Roger Penrose, are still controversial [19]. Is there any connection between physics and religion? Can religion provide some insights that that help solve the difficult problems mentioned above in physics? These are clearly open questions and it is the aim of this paper to bring physics and religion together.

So far there are a lot of discussions in science and religion among scholars, however, none can put into serious and formal theoretical physics construction, and quite many informal researches are merely pseudo-science. Up-to-date, there is only an experimental study that co-relates Buddhism chanting to neuropsychological effect [20]. This paper aims at providing a first, formal, rigorous theoretical ground for Buddhism, and our previous work in [1] has provided the ground of Taoism.

The Chinese philosophy of Yi and Taoism is based on the theory of dualism of yin and yang. The fundamental concepts of opposition and complementation are manifested beautifully in dualism, namely in the complete dual structure we introduced in our previous paper [1].

Buddhism, as another global religion with deep philosophical thought, provides in-depth insights on the full cosmological view beyond the materialistic realm. In particular, amid all sects and schools of Buddhism, the Mādhyamaka school in Mahayana Buddhism established by Nāgārjuna after 500 years of Nirvana of the Gautama Buddha advocates the teaching of the two truths. The two-truths doctrine is dichotomized into two ontological levels, known as the two satya, the conventional truth and provisional truth with opposition to each other. Later, discourses and academic accomplishments of the Mādhyamaka school and Yogachara school divides into the school of emptiness and the school of existence, for which disputes on the fundamental ontological theories of emptiness and existence have been aroused for so many years. In this regard, we would like to apply the theory of duality, or the philosophy thought of Yi and Taoism to resolve the incompatible argument of the two schools. In this chapter, we also aim at presenting the core theory of Heart sutra, Prajnaparamita and Diamond sutra in a formal physics and mathematical way. This also helps to clear the ambiguity of the language used in the sutra.

The paper is divided into three parts. Chapter 2 and Chapter 3 are a review of the essence of the theory of duality that is needed for the theoretical development of Buddhism. The main part of physics of Buddhism is written in chapter 4, which proves several important theorems in Buddhism. Finally, chapter 5 is the concluding remark of the whole paper.

# Chapter 2

# The Theory of Duality and Fundamental Formalism

## 2.1 Single Duality Structure

Since the concept of duality will be used to develop the theory of Buddhism, we will first give a review in this subject matter (p. 4-17) (details of the full subject please refer to [1]). The structure of duality is defined as follow:

**Definition 2.1.1.** (I) Let $U$ be an element set and its dual $U^*$, where there exists a one-to-one bijective map on $U$ and $U^*$. Define the duality map $*$, as a function $* : U \to U^*$ which is a representation of the parity group $\mathbb{Z}_2$. The inverse is just the map itself $* : U^* \to U$. The double duality map $**$ is the identity map $I_d$, that $** : U \to U$ and $** : U^* \to U^*$. $U$ and $U^*$ are said to be dual if $U \cap U^* = \emptyset$ under the $*$ map. Define the zero set as $\{0\}$. The complete duality set $W$ is defined by $W = U \cup U^* \cup \{0\}$. The concept of zero is introduced such that $U = U^*$ is dual invariant if and only if $U = U^* = \{0\}$. For $u_i \in U$ and $u_i^* \in U^*$, there exist a map $\cdot$ such that $u_i \cdot u_i^* = 0$, where 0 here is the zero element in the zero set.

(II) The duality set embedded in an extrinsic observer frame in $k$-dimension is said to be a complete single duality structure. The observer's frame of in $k$ dimension forms a duality $S_k$ and $S_k^\star$. Let the duality operator for observer's frame be a map $\star : S_k \to S_k^*$, which is a is a representation of the parity group $\mathbb{Z}_2$. The inverse is the operator itself, $\star : S_k^\star \to S_k$, and $\star\star$ is the identity map $I_d$. We define the zero set for observer as $\{0\}_{S_k}$. For observer at $\{0\}_{S_k}$, it is defined as the intrinsic observer of the duality system. We concern the extrinsic frame, and define the complete observer's as $B = S_k \cup S_k^\star$. There exists a set in the duality set which is independent of the observer's frame, which is the zero set $\{0\} \in W$. The complete duality structure is defined as $\{W, B\}$.

(III) Each set or dual set have to be observer's frame specific. In the $S_k$ observer's frame, we specify the set and dual set under the observer frame is denoted by $(U|S_k)$ and $(U^*|S_k)$ respectively. In the $S_k^*$ observer's frame, we have $(U|S_k^\star)$ and $(U^*|S_k^\star)$ respectively.

(IV) The dual equivalence of two elements $a, b$, denoted by $a \equiv b$ (or $a :=: b, a* = *b$

) is defined by

$$a \equiv b \,\text{if} \begin{cases} a = b \\ \text{or} \\ a \neq b \text{ but } a, b \,\text{are equivalent by some relation establishment} \end{cases} \tag{2.1}$$

(V) In complete duality, we have the following identity for dual equivalence,

$$(U|S_k) \equiv (U^*|S_k^\star) \quad \text{and} \quad (U|S_k^\star) \equiv (U^*|S_k)\,. \tag{2.2}$$

Element-wise, let $u \in U$ and $u^* \in U^*$, we have

$$(u|S_k) \equiv (u^*|S_k^\star) \quad \text{and} \quad (u|S_k^\star) \equiv (u^*|S_k)\,. \tag{2.3}$$

(VI) The duality operator of the element set $*$ acts as the following:

$$*(u|S_k) = (u^*|S_k) \quad , \quad **\,(u|S_k) = *(u^*|S_k) = (u|S_k)\,. \tag{2.4}$$

(VII) The duality operator of observer set $*$ acts as the following:

$$\star(u|S_k) = (u|S_k^\star) \quad , \quad \star\star\,(u|S_k) = \star(u|S_k^\star) = (u|S_k)\,. \tag{2.5}$$

(VIII) The dual map of the set and the dual operator can act together, which is an identity map. The two different duality map commutes. In other words, $\star \circ * = * \circ \star = I_d$. From example, we have,

$$\star *\,(u|S_k) \equiv \star(u^*|S_k) \equiv (u^*|S_k^\star) \quad \text{and} \quad * \star(u|S_k) = *(u|S_k^\star) = (u^*|S_k^\star)\,. \tag{2.6}$$

(IX) The $\{I_d, *\}$ and $\{I'_d, \star\}$ are elements of two parity groups $\mathbb{Z}_2$ under multiplication. The Klein-4 group, which is the called the 4-duality group, is $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{I, *, \star, * \circ \star\}$. The $(u|S_k), (u^*|S_k), (u|S_k^\star)$ and $(u^*|S_k^\star)$ form a 4-representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$, and can be represented by a 4-tableau diagram,

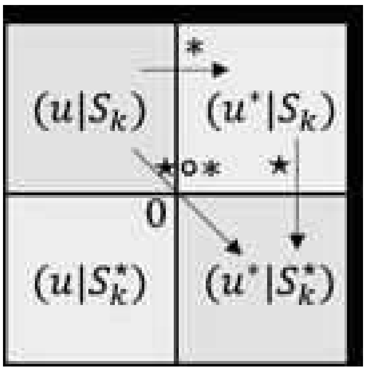


Figure 2.1: The 4-tableau representation of a complete duality structure. Boxes with the same colour denote dual equivalence relation of each other.

The last definition (IX) is achieved by constructing an isomorphism from the four cases to the basis of $\mathbb{Z}_2 \times \mathbb{Z}_2$, such we have a one-to-one map identification as

$$(u|S_k) \to |00\rangle\,, \quad (u|S_k^\star) \to |01\rangle\,, \quad (u^*|S_k) \to |10\rangle\,, \quad (u^*|S_k^\star) \to |11\rangle,. \tag{2.7}$$

The $(u|S_k)$ is called an identity element, in which no dual operation is acted upon on it. Without loss of generality, we can also pick $(u^*|S_k^\star)$ as the identity. In terms of the number of dual operation that act on the element and observer, we can write

$$0+0 \equiv 1+1 \quad \text{and} \quad 0+1 \equiv 1+0\,. \tag{2.8}$$

The element-observer composite $(\,|\,)$ can also be viewed as a tensor product, i.e. $(a|b) \equiv (a| \otimes |b) \equiv a \otimes b$. Consider the basis of representation of $\mathbb{Z}_2$ be $u \oplus u^*$, and the basis of representation of another $\mathbb{Z}_2$ as $S_k \oplus S_k^\star$. Then we have

$$\begin{aligned}(u \oplus u^*) \otimes (S_k \oplus S_k^\star) &= (u \otimes S_k) \oplus (u \otimes S_k^\star) \oplus (u^* \otimes S_k) \oplus (u^* \otimes S_k^\star)\\ &= (u|S_k) \oplus (u|S_k^\star) \oplus (u^*|S_k) \oplus (u^*|S_k^\star)\,,\end{aligned} \tag{2.9}$$

which is the basis of representation of $\mathbb{Z}_2 \otimes \mathbb{Z}_2$ direct product group. But since $\mathbb{Z}_2 \otimes \mathbb{Z}_2 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, therefore the above serves as the basis of the 4-duality group. For simplicity, using 2.7 we can write it as

$$\psi = |00\rangle \oplus |01\rangle \oplus |10\rangle \oplus |11\rangle \tag{2.10}$$

Since by recalling that 2.3, $(u|S_k) \equiv (u^*|S_k^\star)$ and $(u|S_k^\star) \equiv (u^*|S_k)$, we have $|00\rangle \equiv |11\rangle$ and $|01\rangle \equiv |10\rangle$. This can be viewed as

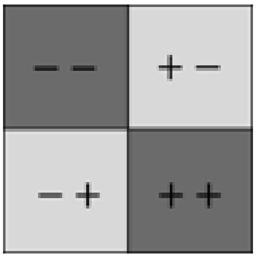


Figure 2.2: The two equivalence relationship can be viewed as $++, -- \to +$ and $+-, -+ \to -$ In other words, considering the backward argument, the conventional relationship of arithmetic can be viewed as the equivalent relationships of tensor product of dual sets as demonstrated above. .

Explicitly in the tensor product representation, we have the operators as

$$\begin{aligned}&(1 \otimes 1)(u \otimes S_k) = (u \otimes S_k)\\ &(* \otimes 1)(u \otimes S_k) = (u^* \otimes S_k)\\ &(1 \otimes \star)(u \otimes S_k) = (u \otimes S_k^\star)\\ &(* \otimes 1)(1 \otimes \star)(u \otimes S_k) = (* \otimes \star)(u \otimes S_k) = (u^* \otimes S_k^\star)\end{aligned} \tag{2.11}$$

We have the Klein-4 group as $\mathbb{Z}_2 \otimes \mathbb{Z}_2 = \{1 \otimes 1, * \otimes 1, 1 \otimes \star, * \otimes \star\}$.

We group the terms as

$$\psi = \big[(u|S_k) \oplus (u^*|S_k^\star)\big]_D \oplus \big[(u|S_k^\star) \oplus (u^*|S_k)\big]_{D^*}\,, \tag{2.12}$$

or

$$\psi = \big[|00\rangle \oplus |11\rangle\big]_D \oplus \big[|01\rangle \oplus |10\rangle\big]_{D^*} = |\psi_D\rangle \oplus |\psi_{D^*}\rangle\,, \tag{2.13}$$

where the subscript $D$ and $D^*$ indicate the two dual partitions, and the two states are orthogonal to each other $\langle\psi_D|\psi_{D^*}\rangle = 0$. Now we will show that in fact the basis of

$\mathbb{Z}_2 \otimes \mathbb{Z}_2$ can be separated into two dual partitions, such that the blue boxes are dual to the green boxes in figure 2.1. Define the parity operator of the partition $D$ as follow,

$$\hat{P}_D = (1 \otimes \star) \oplus (1 \otimes \star) \,. \tag{2.14}$$

Then we have

$$\begin{aligned}
\hat{P}_D \big[|00\rangle \oplus |11\rangle\big]_D &= \big[(1 \otimes \star) \oplus (1 \otimes \star)\big] \big[|00\rangle \oplus |11\rangle\big]_D \\
&= (1 \otimes \star)(|0\rangle \otimes |0\rangle) \oplus (1 \otimes \star)(|1\rangle \otimes |1\rangle) \\
&= (|0\rangle \otimes |1\rangle) \oplus (|1\rangle \otimes |0\rangle) \\
&= \big[|01\rangle \oplus |10\rangle\big]_{D^*} \,.
\end{aligned} \tag{2.15}$$

It follows that

$$\hat{P}_D \big[|01\rangle \oplus |10\rangle\big]_{D^*} = \hat{P}_D^2 |00\rangle \oplus |11\rangle\big]_D = [|00\rangle \oplus [|11\rangle\big]_D \,. \tag{2.16}$$

It can be also easily checked that

$$\begin{aligned}
\hat{P}_D^2 &= [(1 \otimes \star) \oplus (1 \otimes \star)][(1 \otimes \star) \oplus (1 \otimes \star)] \\
&= (1 \otimes \star)(1 \otimes \star) \oplus (1 \otimes \star)(1 \otimes \star) \\
&= (1 \otimes 1) \oplus (1 \otimes 1) \\
&= 1 \oplus 1 \\
&= I
\end{aligned} \tag{2.17}$$

which is the identity matrix. Therefore, the two bases $\big[|00\rangle \oplus |11\rangle\big]_D$ and $\big[|01\rangle \oplus |10\rangle\big]_{D^*}$ are the basis of of the duality group $\mathbb{Z}_2$. Therefore, $\psi$ can be decomposed into two EPR basis pair. Symbolically we can write

$$\psi = 2 \oplus 2 \,. \tag{2.18}$$

Note that the choice of $\hat{P}_D$ is not unique, we can also define,

$$\hat{Q}_D = (* \otimes 1) \oplus (* \otimes 1) \,. \tag{2.19}$$

Then we have

$$\begin{aligned}
\hat{Q}_D \big[|00\rangle \oplus |11\rangle\big]_D &= \big[(* \otimes 1) \oplus (* \otimes 1)\big] \big[|00\rangle \oplus |11\rangle\big]_D \\
&= (* \otimes 1)(|0\rangle \otimes |0\rangle) \oplus (* \otimes 1)(|1\rangle \otimes |1\rangle) \\
&= (|1\rangle\rangle \otimes |0\rangle\rangle) \oplus (|0\rangle \otimes |1\rangle) \\
&= \big[|01\rangle \oplus |10\rangle\big]_{D^*} \,.
\end{aligned} \tag{2.20}$$

And similarly we have $\hat{Q}_D^2 = I$ which is the identity map.

Furthermore, we can have rectangular duality. We consider $\psi$ with the following partitions,

$$\psi = \big[|00\rangle \oplus |01\rangle\big]_P \oplus \big[|11\rangle \oplus |10\rangle\big]_{P^*} = |\psi_P\rangle \oplus |\psi_{P^*}\rangle \,. \tag{2.21}$$

Clearly partitions $P$ and $P^*$ are dual to each other, and the two dual basis are orthogonal to each other $\langle \psi_P | \psi_{p^*} \rangle = 0$. This is referred as the the vertical rectangular duality. Similarly, we can have

$$\psi = \big[|00\rangle \oplus |10\rangle\big]_Q \oplus \big[|11\rangle \oplus |01\rangle\big]_{Q^*} = |\psi_Q\rangle \oplus |\psi_{Q^*}\rangle \,, \tag{2.22}$$

where $Q$ and $Q^*$ are dual to each other, and the two dual basis are orthogonal to each other $\langle\psi_Q|\psi_{Q^*}\rangle = 0$. This is referred as the horizontal rectangular duality. The idea is illustrated as follow.

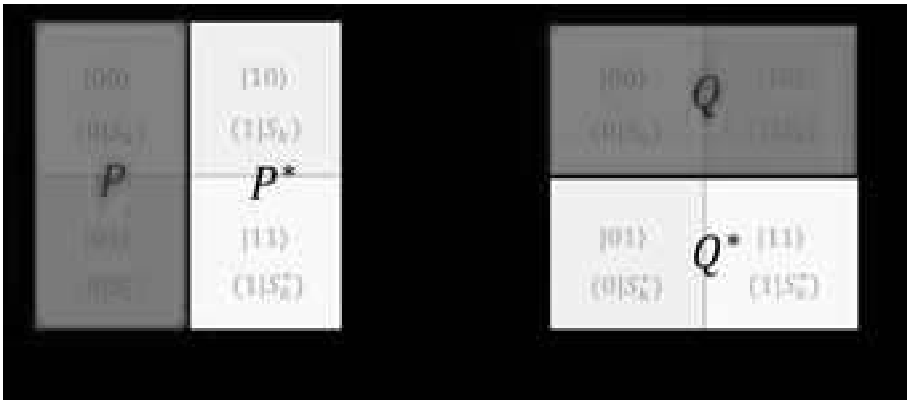


Figure 2.3

Explicitly, we can draw out the whole idea of duality for illustration. The dual elements and dual observers form a generic 4-dual diagram as follow.

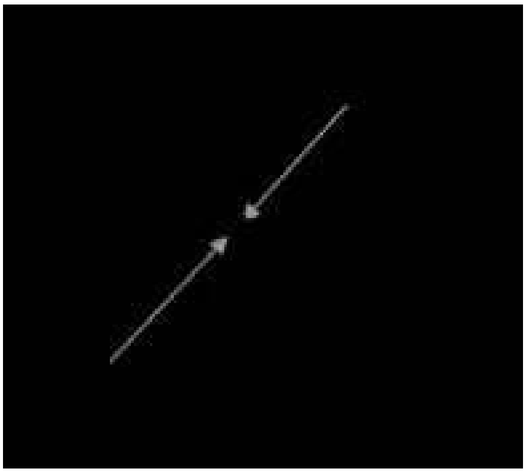

Figure 2.4: In the diagram, we can see that $U$ observed by $S_k$ is equivalent to $U^*$ observed by $S_k^\star$ such that $(U|S_k) \equiv (U^*|S_k^\star)$; and $U^*$ observed by $S_k$ is equivalent to $U$ observed by $S_k^\star$ such that $(U^*|S_k) \equiv (U|S_k^\star)$.

The above abstract definition can be easily understood by some examples. The simplest case for a dual system would be positive and negative numbers. Let start from the most fundamental case. Let $U = \{-1\}$ and $U^* = \{+1\}$, and the zero set $\{0\}$. Consider a pair of dual observers living on a 2D manifold, the one in front of the two numbers is $S_2$, and the one behind the two numbers is $S_2^*$. Let's use the normal convention of a number line, the left is -1 and the right is +1, then we have,

$$(-1|S_2) \equiv (+1|S_2^*) \quad \text{and} \quad (-1|S_2^*) \equiv (+1|S_2)\,. \tag{2.23}$$

And we have the map $\cdot = +$ such that $(-1)+(+1) = 0$. If we let $U = \mathbb{R}^-$ and $U^* = \mathbb{R}^+$ and the zero set, the you have the duality for the real number system. If you still find it abstract, you can consider the case of the mirror. Left becomes right in the mirror's frame, and vice versa. This is exactly the concept we demonstrate. Another example would be spin. Let $U = \{\downarrow\} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $U^* = \{\uparrow\} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ with $** = \mathbb{I}$, where we observe in a 3 dimensional space. Then we have

$$(\downarrow |S_3) \equiv (\uparrow |S_3^*) \quad \text{and} \quad (\downarrow |S_3^*) \equiv (\uparrow |S_3)\,. \tag{2.24}$$

This is demonstrated in figure 2.5. And we have the $\cdot$ map as the inner product $\langle \uparrow | \downarrow \rangle = 0$.

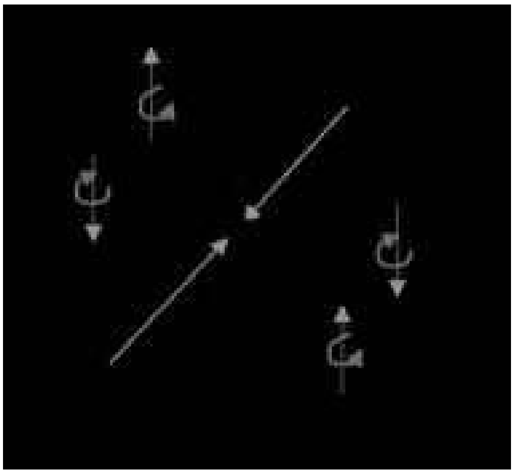

Figure 2.5

Although not obviously noticed, spontaneously duality symmetry breaking of choice is always implicitly inferred. For example we define left-hand side as negative in our observation perspective, but this is equivalently to a positive right-hand side in the dual perspective. However we often make a particular choice of representation so that at the end only one representation out of the two equivalence is used. Without the loss of generality we can pick the dual one, but for realistic observable we must pick a particular one. In quantum mechanics terms, this is a state collapse of a dual state. Explicitly,

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\Big(|(U, U^*|S_k)\rangle + |(U^*, U|S_k^\star)\rangle\Big)\,, \tag{2.25}$$

where we can map $|(U, U^*|S_k)\rangle \to |01\rangle$ and $|(U^*, U|S_k^\star\rangle \to |10\rangle)$ respectively, with equal probability of $1/2$. This is an EPR pair and an entangled state. In general we can write

$$|\Psi\rangle = \cos\theta|(U, U^*|S_k)\rangle + \sin\theta|(U^*, U|S_k^\star\rangle\,, \tag{2.26}$$

When the phase is at $\pi/4$, we have both the probabilities as $1/2$. At $\theta = 0$ or $2\pi$, we have a deterministic state for $|(U, U^*|S_k)\rangle$ and at $\theta = \pi$, we have a deterministic state for $|(U^*, U|S_k^\star)\rangle$.

Next we would like to promote the idea into a more abstract way. We can call $U$ as a left dual ${}^*U$. In figure 2.4, if we slice along the $S_k, S_k^\star$ frame, we can see the pair of element $U^{**}U$ and its dual ${}^*UU^*$. We identify as follow:

$$U^{**}U \text{as } RL \quad \text{and} \quad {}^*UU^* \text{as } LR\,. \tag{2.27}$$

In the case of of RL, the two $*$s are in the inner side and we term this as “bonding” denoted as $\rightarrow\leftarrow$, while in the case of LR, the two $*$s are at the outer side and we term this as “anti-bonding” denoted as $\longleftrightarrow$. Thus the ”bonding” and ”anti-bonding” representation is a dual representation. And the two objects $U^{**}U$ and $^{*}UU^{*}$ form a basis of irreducible representation of $\mathbb{Z}_2$. We can go in the other way that if there exist such a dual pair, then the notion of observation frame is implied.

The role of element and observer is interchangeable in a 4-duality system. Now we can treat the element as observer and observer as element, this is known as element-observer duality.

### 2.1.1 The dual equation

Finally we would like to write out the duality theory in a compact form. Let $a = (u|S_k)$, and the full negation operator be $! = \star *$, then we nicely obtain the follow equation,

$$a \equiv !a\,. \tag{2.28}$$

The negation of this equation is

$$!a \equiv \ a\,. \tag{2.29}$$

This is because

$$!! = \star * \star * = (\star \otimes *)(\star \otimes *) = (\star\star \otimes **) = (\mathbf{1} \otimes \mathbf{1}) = \mathbf{1}. \tag{2.30}$$

Equation 2.28 is the same as 2.28. Thus both equations 2.28 and 2.29 are dual invariant. On the other hand, let $\bar{a} = (u^*|S_k)$ then we have

$$\bar{a} \equiv !\bar{a}\,. \tag{2.31}$$

The negation of this equation is

$$!\bar{a} \equiv \ a \tag{2.32}$$

which is same as 2.31. Thus both equations 2.31 and 2.32 are dual invariant. Therefore, the dual equation consists of 4 equations 2.28, 2.29, 2.31 , 2.32,

$$\begin{cases} a \equiv !a \text{ and } !a \equiv \ a \text{ for partition } D \\ \bar{a} \equiv !\bar{a} \text{ and } !\bar{a} \equiv \ \bar{a} \text{ for partition } D^* \end{cases}. \tag{2.33}$$

Diagramatically,

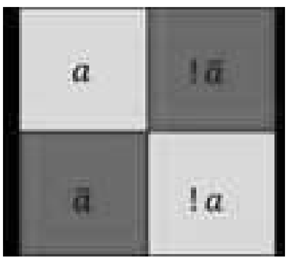


Figure 2.6

### 2.1.2 Representation of dual operators

In the section we will construct matrix representations of the dual operators. Let $|0\rangle$ and $|1\rangle$ be a pair of dual states, satisfying $*|0\rangle = |1\rangle$ and $*|1\rangle = |0\rangle$ where $* = \hat{P}$ is the dual operator, satisfying the orthogonal relation $\langle 0|1\rangle = \langle 1|0\rangle = 0$.

The identity element can be constructed by

$$\mathbf{1} = |0\rangle\langle 0| + |1\rangle\langle 1| \,, \tag{2.34}$$

and the parity element can be constructed by

$$\hat{P} = |0\rangle\langle 1| + |1\rangle\langle 0| \,. \tag{2.35}$$

This can be proven easily by

$$\begin{aligned}
\hat{P}^2 &= (|0\rangle\langle 1| + |1\rangle\langle 0|)(|0\rangle\langle 1| + |1\rangle\langle 0|) \\
&= |0\rangle\langle 1|0\rangle\langle 1| + |0\rangle\langle 1|1\rangle\langle 0| + |1\rangle\langle 0|0\rangle\langle 1| + |1\rangle\langle 0|1\rangle\langle 0| \\
&= |0\rangle\langle 0| + |1\rangle\langle 1| \\
&= \mathbf{1} \,.
\end{aligned} \tag{2.36}$$

It can also be checked that

$$\begin{aligned}
\mathbf{1}^2 &= (|0\rangle\langle 0| + |1\rangle\langle 1|)(|0\rangle\langle 0| + |1\rangle\langle 1|) \\
&= |0\rangle\langle 0|0\rangle\langle 0| + |0\rangle\langle 0|1\rangle\langle 1| + |1\rangle\langle 1|0\rangle\langle 0| + |1\rangle\langle 1|1\rangle\langle 1| \\
&= |0\rangle\langle 0| + |1\rangle\langle 1| \\
&= \mathbf{1} \,.
\end{aligned} \tag{2.37}$$

Next, to construct matrix representation for $\mathbf{1}$ and $\hat{P}$,

$$\mathbf{1}_{ab} = \langle a|\mathbf{1}|b\rangle \,. \tag{2.38}$$

Explicitly,

$$\mathbf{1} = \begin{pmatrix} \langle 0|\mathbf{1}|0\rangle & \langle 0|\mathbf{1}|1\rangle \\ \langle 1|\mathbf{1}|0\rangle & \langle 1|\mathbf{1}|1\rangle \end{pmatrix} \,. \tag{2.39}$$

To evaluate the matrix element, we compute

$$\begin{aligned}
\mathbf{1}_{ab} &= \langle a|0\rangle\langle 0|b\rangle + \langle a|1\rangle\langle 1|b\rangle \\
&= \delta_{a0}\delta_{0b} = \delta_{a1}\delta_{1b} \,.
\end{aligned} \tag{2.40}$$

Therefore as expected,

$$\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \,. \tag{2.41}$$

For $\hat{P}$,

$$\hat{P}_{ab} = \langle a|\hat{P}|b\rangle \,. \tag{2.42}$$

Explicitly,

$$\hat{P} = \begin{pmatrix} \langle 0|\hat{P}|0\rangle & \langle 0|\hat{P}|1\rangle \\ \langle 1|\hat{P}|0\rangle & \langle 1|\hat{P}|1\rangle \end{pmatrix} \,. \tag{2.43}$$

To evaluate the matrix element,

$$\begin{aligned}\hat{P}_{ab} &= \langle a|0\rangle\langle 1|b\rangle + \langle a|1\rangle\langle 0|b\rangle \\ &= \delta_{a0}\delta_{1b} + \delta_{a1}\delta_{0b}\,.\end{aligned} \tag{2.44}$$

Therefore we have

$$\hat{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\,. \tag{2.45}$$

which is called the $\boldsymbol{M}$ matrix. The matrix representation in 2.45 can be diagonalized, since the eigenvalue is 1 or $-1$, we have

$$\hat{P}' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\,. \tag{2.46}$$

which is another representation of the parity operator. And there exists a similarity transformation such that $U\hat{P}U^{-1} = \hat{P}'$.

One can observe that because

$$\mathbf{1}\cdot\mathbf{1} = \mathbf{1} \quad , \quad \hat{P}\cdot\hat{P} = \mathbf{1} \quad , \quad \mathbf{1}\cdot\hat{P} = \hat{P} \quad , \quad \hat{P}\cdot\mathbf{1} = \hat{P}\,, \tag{2.47}$$

this is isomorphic to the law of arithmetic multiplication that $++ = +, -- = +, +- = -, -+ = -$. Thus we can identify $+ \to \mathbf{1}$, $- \to \hat{P}$.

# Chapter 3

# Construction of the diagramatic basis representation of 4-duality group

In this chapter we study the construction of basis of irreducible representation of the 4-duality group $\mathbb{Z}_2 \times \mathbb{Z}_2$. We would extensively use the diagramatic representation of the 4-box tableaux, which is called the 4-fundamental tableaux representation,

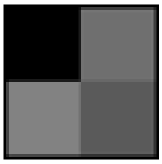

Figure 3.1: 4-box tableaux representation of the 4-duality group

Recalling the definition of the dual space, which consists of two element vector space of $V$ and $V^*$ which are isomorphic to each other [1]. Here each coloured box, (red (R), blue (B), magenta (M) and white (W) ) represent the following,

$$\mathrm{R} := U, \quad \mathrm{B} := U^*, \quad \mathrm{M} := \mathrm{U} \cup \mathrm{U}^*, \quad \mathrm{W} = \mathrm{U} \cap \mathrm{U}^* = \emptyset \,. \tag{3.1}$$

And in particular, we have $W = M^* = *(U \cup U^*) = U^* \cap U = \emptyset$ and $*\emptyset = *(U^* \cap U) = U \cup U^* = W$, where we have used de-Morgan's theorem. Thus $W$ and $M$ is dual to each other. The full union is sometimes written as 'All' , while the null intersection is sometimes written as 'Null' or 'None'. This can be understood diagramatically by the four colour in the 4-fundamental tableaux representation. The magenta is the mixing of red and blue, while the white has no overlap between them. The origin, which is defined as the central zero, can be omitted at the moment. We also define each coloured box to have unity unit of area, thus the 4-fundamental tableau is a 4-unit object. Next we define the 4 quadrants for the 4-fundamental tableaux representation. The four quadrants correspond to the 4 boxes, for which each quadrant is a vector space $Q$. The quadrant number is defined by indexing the coloured boxes by the following binary number $q_Q$:

$$\mathrm{W} : (00),\, 0; \quad \mathrm{R} : (01),\, 1; \quad \mathrm{B} : (10),\, 2; \quad \mathrm{M} : (11), 3\,. \tag{3.2}$$

[1]In the most general general definition of dual space the isomorphism is not necessarily imposed.

In terms of vectors,

$$\mathrm{R} := v, \quad \mathrm{B} := v^*, \quad \mathrm{M} := v + v^*, \quad W = \mathbf{0}\,, \tag{3.3}$$

Suppose $|0\rangle \in V$ and $|1\rangle \in V^*$ where $V$ and $V^*$ are one dimensional vector spaces. Now consider the qubit $|\psi(\theta)\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle$. For convenience, we consider the labelling of the zero vector,

$$0|u\rangle = \mathbf{0}_u = \mathbf{0}\,, \quad 0|u^*\rangle = \mathbf{0}_{u^*} = \mathbf{0}\,. \tag{3.4}$$

We also take

$$*\mathbf{0}_u = \mathbf{0}_{u^*}\,, \quad *\mathbf{0}_{u^*} = \mathbf{0}_u \tag{3.5}$$

But note that the zero vector itself has no parity, we just introduce it for convenience, so notice that

$$\mathbf{0}_u = \mathbf{0}_{u^*} = \mathbf{0}\,. \tag{3.6}$$

Then we have

$$|\psi(0)\rangle = |0\rangle + 0|1\rangle = |0\rangle + \mathbf{0}_1\,, \tag{3.7}$$

$$|\psi(\pi/2)\rangle = 0|0\rangle + |1\rangle = \mathbf{0}_0 + |1\rangle\,, \tag{3.8}$$

The fundamental four-box tableaux now reads

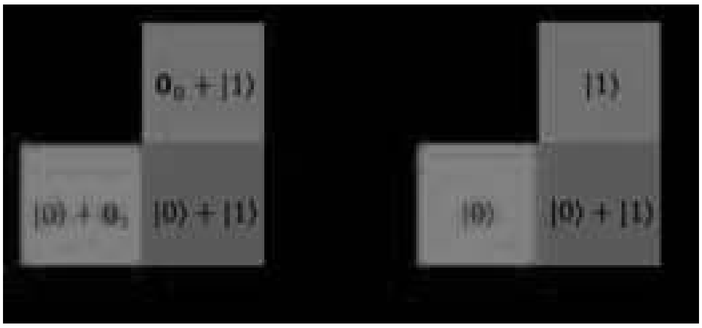


Figure 3.2

We can see that

$$|\psi(\pi/2)\rangle = *|\psi(0)\rangle \tag{3.9}$$

The Null state and the All state are dual invariant,

$$*\mathbf{0} = \mathbf{0}_1 + \mathbf{0}_0 = \mathbf{0} \quad \text{and} \quad *\left(|0\rangle + |1\rangle\right) = |1\rangle + |0\rangle = |0\rangle + |1\rangle\,, \tag{3.10}$$

while the half states are not. We can further check that states in the dual partition are orthogonal. For R and B, we have $\langle v|v^*\rangle = \langle 0|1\rangle = 0$ which satisfies. So clearly they are orthogonal. For M and W, it is clearly that $\langle \mathbf{0}|(|0\rangle + |1\rangle) = 0$ so the full vector and the null vector are orthogonal and dual to each other. Notice that the role of zero vector is to ensure $|v\rangle + \mathbf{0} = \mathbf{0} + |v\rangle = |v\rangle$, this is in analogy to the role of empty set that $U \cup \emptyset = \emptyset \cup U = U$.

Then in dimensional representation we have

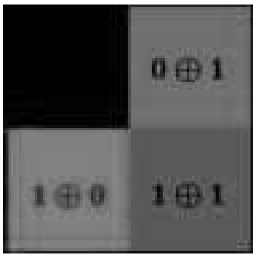


Figure 3.3

The dimensional representation is isomorphic to the basis of $\mathbb{Z}_2 \times \mathbb{Z}_2$. Thus we have constructed the basis representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ by the dimensional representation of the 4-box tableaux.

Now we would like to construct another new representation in terms of direct-sum vector space.

$$\mathrm{R} := V \oplus \mathbf{0}_{V^*}, \quad \mathrm{B} := \mathbf{0}_V \oplus V^*, \quad \mathrm{M} := V \oplus V^*, \quad W = \mathbf{0}_V \oplus \mathbf{0}_{V^*}, \tag{3.11}$$

where

$$\mathbf{0}_V = \mathbf{0}_{V^*} = \mathbf{0} = |\mathbf{0}\rangle\,. \tag{3.12}$$

In terms of element,

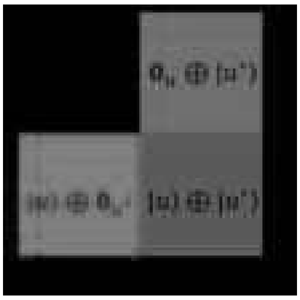

Figure 3.4

The zero vector is the only dual invariant element in the representation, such that

$$*\mathbf{0} = \mathbf{0}\,. \tag{3.13}$$

Next we can check that the vectors in the same dual partition are orthogonal. We need to use the following identity,

$$\left\langle \bigoplus_{i=1}^{n} v_i \middle| \bigoplus_{i=1}^{n} u_i \right\rangle = \sum_{i=1}^{n} \langle v_i | u_i \rangle\,. \tag{3.14}$$

Let $|w_{\mathrm{full}}\rangle = |u\rangle \oplus |u^*\rangle$, $|w_+\rangle = |u\rangle \oplus |\mathbf{0}_{u^*}\rangle$, $|w_-\rangle = |\mathbf{0}_u\rangle \oplus |u^*\rangle$ and $|w_{\mathrm{null}}\rangle = \mathbf{0}$, we have

$$\begin{aligned} \langle w_{\mathrm{null}} | w_{\mathrm{full}} \rangle &= \langle \mathbf{0} | u \oplus u^* \rangle = \langle \mathbf{0}_u | u \rangle + \langle \mathbf{0}_{u^*} | u^* \rangle = 0 \\ \langle w_+ | w_- \rangle &= \langle u \oplus \mathbf{0}_{u^*} | \mathbf{0}_u \oplus u^* \rangle = \langle u | \mathbf{0}_u \rangle + \langle \mathbf{0}_{u^*} | u^* \rangle = 0\,. \end{aligned} \tag{3.15}$$

Furthermore, we can check that we can generate dual vector simply by acting the $*$ on the vector that satisfies orthogonality. The dual construction of 3.4 is illustrated as follows in 3.5:

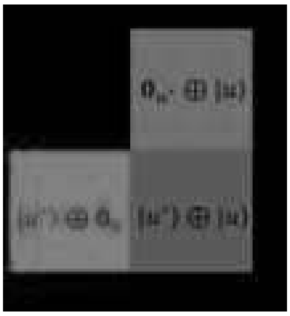


Figure 3.5

We can check that,

$$\begin{aligned}
&\langle w_{\rm full}^* | w_{\rm full}\rangle = \langle u^* \oplus u | u \oplus u^*\rangle = \langle u^*|u\rangle + \langle u|u^*\rangle = 0\,.\\
&\langle w_+^* | w_+\rangle = \langle u^* \oplus \mathbf{0}_u | u \oplus \mathbf{0}_{u^*}\rangle = \langle u^*|u\rangle + \langle \mathbf{0}_u|\mathbf{0}_{u^*}\rangle = 0\,.\\
&\langle w_-^* | w_-\rangle = \langle \mathbf{0}_{u^*} \oplus u | \mathbf{0}_u \oplus u^*\rangle = \langle \mathbf{0}_{u^*}|\mathbf{0}_u\rangle + \langle u|u^*\rangle = 0\,.\\
&\langle w_{\rm null}^* | w_{\rm null}\rangle = \langle \mathbf{0}|\mathbf{0}\rangle = 0\,.
\end{aligned} \tag{3.16}$$

# Chapter 4

# The Physics of Buddhism

## 4.1 Introduction to the Mādhyamaka school of Buddhism and Prajnaparamita

The concept of emptiness appears throughout the Buddhist thoughts and texts. The mundane view of emptiness is nothingness, which is considered as the negation of existence (being). However, in Buddhism, the notion of emptiness does not refer to nothingness, in fact it refers to the absence of an independent nature. This is known as the Prajnaparamita. A phenomenon is caused by many conditions and inter-be with other components, for which it lacks the independent existing entity, and thus considered as empty in the Buddhist view. Ordinary sentient beings, who are unenlightened, are always in attachment to existence of being, but lack the Prajnaparamita perspective to view the emptiness perspective of all phenomena. The existence of being considered by sentient beings are considered as empty by the Buddha (or sages, Buddha's disciples), and vice versa. This statement can already be written down mathematically by our 4-dual structure in equation 2.3,

$$(\text{being} \,|\, \text{ordinary senting beings}) \equiv (\text{emptiness} \,|\, \text{Buddha})\,, \tag{4.1}$$

$$(\text{emptiness} \,|\, \text{ordinary senting beings}) \equiv (\text{being} \,|\, \text{Buddha})\,. \tag{4.2}$$

Here we take the dual set $U = \{\text{being}\}$ and $U^* = \{\text{emptiness}\}$, and the perspective as $S_k$ = Ordinary senting beings and its dual $S_k^\star = \{\text{Buddha}\}$. We have $*\text{being} =$ emptiness and $\star$(Ordinary senting beings) = Buddha.

We can also inter mathematically the being element as $|1\rangle$ state and the emptiness element as $|0\rangle$ state, for which they are orthogonal

$$\langle \text{being}|\text{emptiness}\rangle = \langle \text{emptiness}|\text{being}\rangle = 0\,. \tag{4.3}$$

The similarity goes to the observation perspective. We can infer mathematically $|1\rangle$ state as the ordinary sentient beings and $|0\rangle$ state be the Buddha. Then we have

$$\langle \text{ordinary sentint beings}|\text{Buddha}\rangle = \langle \text{Buddha}|\text{ordinary sentint beings}\rangle = 0\,. \tag{4.4}$$

For convenience, let's symbolize the above wordy expressions into mathematics. From now on, we denote Buddha as $\mathcal{B}$, ordinary sentient being as $\mathcal{S}$. We simply have $\star\mathcal{B} = \mathcal{S}$ and $\star\mathcal{S} = \mathcal{B}$ with $\star\star = 1$. Then we have $\langle\mathcal{S}|\mathcal{B}\rangle = \langle\mathcal{B}|\mathcal{S}\rangle = 0$. Next we denote being as a black square ■, and emptiness as an empty square □. We simply have $*■ = □$ and $*□ = ■$ with $** = 1$. We will use these notations throughout the text in order to save the clumsiness of wordy expressions. The complete state will then be

$$\psi = (■|\mathcal{S}) \oplus (□|\mathcal{B}) \oplus (■|\mathcal{B}) \oplus (□|\mathcal{S})\,. \tag{4.5}$$

This is nothing but just,

$$\psi = (1|1) \oplus (0|0) \oplus (1|0) \oplus (0|1)\,. \tag{4.6}$$

in a familiar sense.

Now we define a dual set of being and nothing. Note that we should not confuse with existence and being, although they both mean presence of something, they are slightly difference here as they they belong to different dual set. Also nothing is different from emptiness as we have explained previously. We simply refer existence as 1 and nothing as 0 as a standard. Next we introduce the concept of actualization and illusion, which is a duality. In Buddhism, we always talk about the illusion of an existence. Ordinary sentient beings reckon that existence is real however in the Buddhist view is empty because it does not have a self-independent nature, so the existence is an illusionary existence, but not an actual existence. We define actualization and illusion as dual operators which act upon the state of existence and emptiness. Denote the actualization operation set as $\hat{O} = \{\$\}$, and the illusion operation set as $\hat{O}^* = \{!\}$. These operators operate as follow,

$$\$|u\rangle = |u\rangle, \quad \$|u^*\rangle = |u^*\rangle, \quad !|u\rangle = |u^*\rangle, \quad !|u^*\rangle = |u\rangle\,. \tag{4.7}$$

So basically \$ is just the identity operator while ! is just the parity operator. Now we take $|u\rangle = |1\rangle$ for the state of being and $|u^*\rangle = |0\rangle$ for the state of nothing. These four operations form the basis of $\mathbb{Z}_2 \times \mathbb{Z}_2$, as it is easy to see that

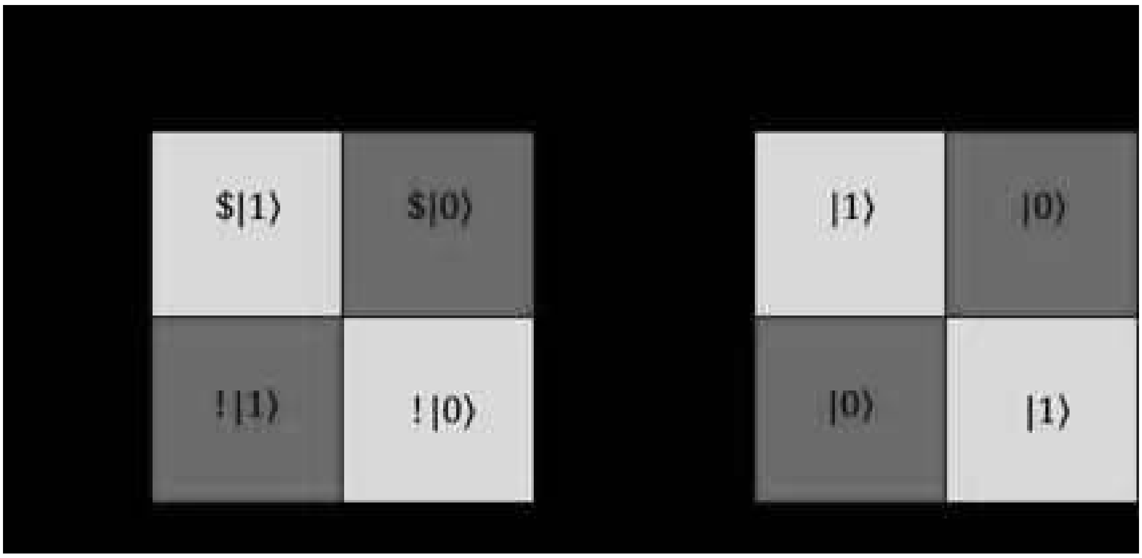


Figure 4.1

for which the isomorphism holds,

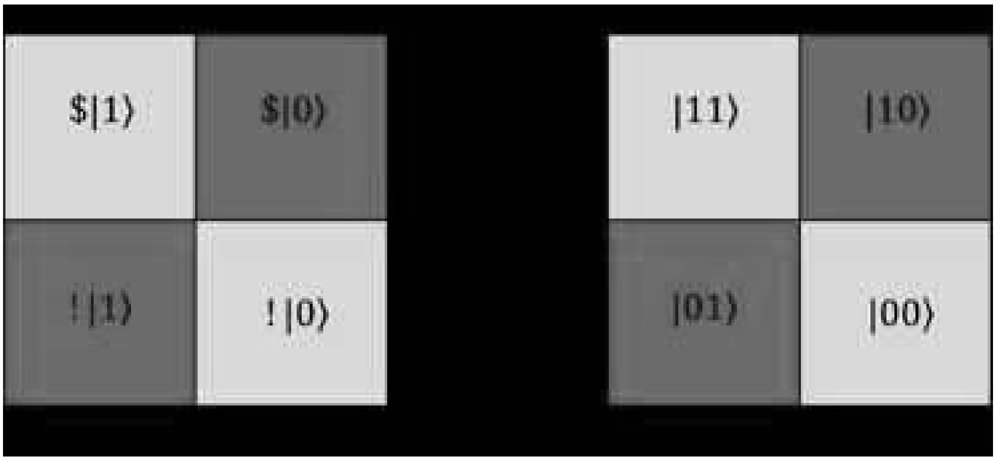


Figure 4.2

The idea is fairly abstract and we will explain here in details. First for $\$|1\rangle$, it means actual being. Actual being is just being $|1\rangle$ because the actualization is real. Second, for $\$|1\rangle$, it means actual nothing. Actual nothing is just nothing $|0\rangle$ because the actualization is real. Then for $!|1\rangle$, it means illusive being. This means nothing $|0\rangle$ as the being is an illusion because it has no independence of existence. Finally $!|0\rangle$ means the illusive nothing. This means being $|1\rangle$ as the nothing is an illusion. To sum up we have

$$\$|1\rangle = !|0\rangle = |1\rangle \quad \text{for the green box}\,, \tag{4.8}$$

$$\$|0\rangle = !|1\rangle = |0\rangle \quad \text{for the blue box}\,. \tag{4.9}$$

Finally notice that

$$*\$|1\rangle = !|1\rangle\,. \tag{4.10}$$

we have

$$*\$ = !\,. \tag{4.11}$$

This means non-actualization is illusion. Taking $*$ both sides on $*\$ = !$, we have

$$**\$ = *! \quad \Longrightarrow \quad \$ = *!\,. \tag{4.12}$$

This means actualization is non-illusion.

### 4.1.1 The canonical 4-duality structure.

In this section, we will establish the canonical 4-duality structure, this formalism appears quite often in the Mahayana Buddhist theory of emptiness. Define a dual basis set $\{|u\rangle, |u^*\rangle\}$ and an dual operator set $\{\hat{e}, \star\hat{e}\}$. The $\hat{e}$ is the 'is' operator and the $\star\hat{e}$ is the 'not' operator. The full state of the canonical 4-duality structure is defined by

$$\psi = \sum_{i=0,1}^{\oplus} \sum_{j=0,1}^{\oplus} (\hat{P}_i \otimes \hat{P}_j)(|u\rangle \otimes |u^*\rangle)\,, \tag{4.13}$$

where $\hat{P}_0 = \hat{e}$ and $\hat{P}_1 = \star\hat{e}$. Explicitly, the sum is

$$\psi = (\hat{e}|u\rangle \otimes \hat{e}|u^*\rangle) \oplus (\hat{e}|u\rangle \otimes \star\hat{e}|u^*\rangle) \oplus (\star\hat{e}|u\rangle \otimes e|u^*\rangle) \oplus (\star\hat{e}|u\rangle \otimes \star\hat{e}|u^*\rangle)\,. \tag{4.14}$$

In terms of a 4-tableau, we have

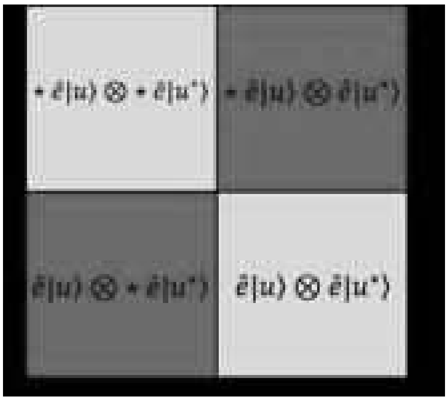


Figure 4.3

It is noted that if the operators operate on the states as follow,

$$\hat{e}|u\rangle = |u\rangle, \quad \hat{e}|u^*\rangle = |u^*\rangle, \quad \star\hat{e}|u\rangle = |u^*\rangle, \quad \star\hat{e}|u^*\rangle = |u\rangle. \tag{4.15}$$

Then we will obtain

$$\psi = |uu^*\rangle \oplus |uu\rangle \oplus |u^*u^*\rangle \oplus |u^*u\rangle. \tag{4.16}$$

This is just the normal full basis. Now we apply the 4-duality structure to study the Buddhist theory. Take $u = \blacksquare$ for existence, $u^* = \square$, then we have

$$\begin{aligned} \psi &= \sum_{i=0,1}^{\oplus} \sum_{j=0,1}^{\oplus} (\hat{P}_i \otimes \hat{P}_j)(|\blacksquare\rangle \otimes |\square\rangle) \\ &= (\hat{e}|\blacksquare\rangle \otimes \hat{e}|\square\rangle) \oplus (\hat{e}|\blacksquare\rangle \otimes \star\hat{e}|\square\rangle) \oplus (\star\hat{e}|\blacksquare\rangle \otimes \hat{e}|\square\rangle) \oplus (\star\hat{e}|\blacksquare\rangle \otimes \star\hat{e}|\square\rangle)\,. \end{aligned} \tag{4.17}$$

In words, this actually means the following:


Figure 4.4

The negation of $\psi$ is given by

$$\begin{aligned} *\psi &= \sum_{i=0,1}^{\oplus} \sum_{j=0,1}^{\oplus} (* \otimes *)(\hat{P}_i \otimes \hat{P}_j)(|\blacksquare\rangle \otimes |\square\rangle) = \sum_{i=0,1}^{\oplus} \sum_{j=0,1}^{\oplus} (*\hat{P}_i \otimes *\hat{P}_j)(|\blacksquare\rangle \otimes |\square\rangle) \\ &= (*\hat{e}|\blacksquare\rangle \otimes *\hat{e}|\square\rangle) \oplus (*\hat{e}|\blacksquare\rangle \otimes * \star \hat{e}|\square\rangle) \oplus (* \star \hat{e}|\blacksquare\rangle \otimes *\hat{e}|\square\rangle) \oplus (* \star \hat{e}|\blacksquare\rangle \otimes * \star \hat{e}|\square\rangle) \end{aligned} \tag{4.18}$$

The $\{\psi, *\psi\}$ is a dual set. It is remarked that the negation operator $*$ is not the same as the not operator $\star$. Only if we want dual invariance $*\psi = \psi$ do we have $* = \star$. In words, this actually means the following:

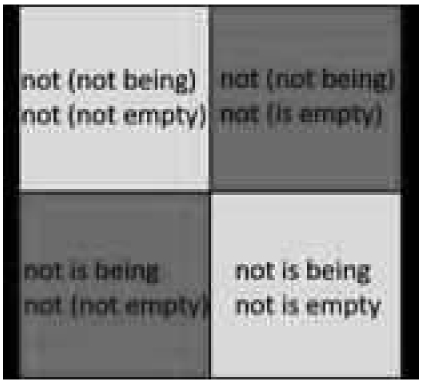


Figure 4.5

It has to be careful that the 'not (not empty)' is not empty. This is not the case of double negation. The first not is from the $*$ operator, while the second not is from the $\star$ operator. The same idea goes to 'not (not being)' is not being. Therefore there is an ambiguity when expressed in words. Another example is, from the Buddhist text there is a heaven called non-thought not non-thought. The 'not non' is not a double negation. Otherwise it is just called the heaven of thought non-thought. Another important property is that the two not operators cannot be exchanged in position, i.e. they are not abelian. Mathematically this means

$$*_1 *_2 |u\rangle \neq *_2 *_1 |u\rangle \,. \tag{4.19}$$

Only if $*_1 = *_2$ we can have $*_1^2 = *_2^2 = I$. The reason is the following. First consider $*_1(*_2|u\rangle)$. Let $a_2 = *_2|u\rangle$, then $*_1(*_2|u\rangle) = *_1 a_2$. Now consider $*_2(*_1|u\rangle)$. Let $a_1 = *_1|u\rangle$, then $*_2 *_1 |u\rangle = *_2 a_1$. Clearly $*_1 a_2 \neq *_2 a_1$ unless $a_2 = a_1$ and $*_1 = *_2$. For example, not anti-matter is obviously different from anti not-matter. Not anti-matter can be photons, apples etc, but anti not-matter refers to the opposite form of non-matter, this excludes apples as apples are matter.

The canonical structure can also be defined through direct sum instead of tensor product as follow,

$$\psi = \sum_{i=0,1}^{\oplus} \sum_{j=0,1}^{\oplus} (\hat{P}_i \oplus \hat{P}_j)(|u\rangle \oplus |u^*\rangle) \,, \tag{4.20}$$

where $\hat{P}_0 = \hat{e}$ and $\hat{P}_1 = \star\hat{e}$. Explicitly, the sum is

$$\psi = (\hat{e}|u\rangle \oplus \hat{e}|u^*\rangle) \oplus (\hat{e}|u\rangle \oplus \star\hat{e}|u^*\rangle) \oplus (\star\hat{e}|u\rangle \oplus e|u^*\rangle) \oplus (\star\hat{e}|u\rangle \oplus \star\hat{e}|u^*\rangle) \,. \tag{4.21}$$

All the remaining properties are the same with the above tensor product case.

### 4.1.2 Form is Emptiness, Emptiness is Form

In this section, we will prove the famously known statement in Prajnaparamita, form is emptiness and emptiness is form which is stated in the Heart Sutra. To be explicit, in line with the rigorous mathematical sense, the correct statement should be: form is equivalent to emptiness and emptiness is equivalent form. This is because 'is' is meant to be $=$, while 'is equivalent to' is meant to be $\equiv$. We should use $\equiv$.

In Buddhism, the existence of interdependence or interconnectedness (in chinese yin yuan) means to be emptying self-independence. The dharma which relies on all other conditions that inter-depend on each other is called the conditioned dharma. This forms the phenomenon of the universe, which is known as the form. Consider the dual set of presence and absence as

$$U = \{\text{being}\} = \{1\} \quad \text{and} \quad U^* = \{\text{absence}\} = \{0\}\,. \tag{4.22}$$

and the observer perspective as

$$S_k = \{\text{interconnectedness}\} \quad \text{and} \quad S_k^\star = \{\text{self}\}\,. \tag{4.23}$$

Using the formula of

$$(U|S_k) \equiv (U^*|S_k^\star)\,, \tag{4.24}$$

we obtain

$$(1|\text{interconnectedness}) \equiv (0|\text{self})\,. \tag{4.25}$$

This is the statement of

$$\boxed{\text{dependent arising } \equiv \text{absent of self}}\,. \tag{4.26}$$

As form is formed by the existence of interconnectedness, we have

$$a = \text{form} = (1|\text{interconnectedness})\,. \tag{4.27}$$

The absence of self (0|self) is referred as emptiness. Therefore by 4.25, we obtain

$$\boxed{\text{form } \equiv \text{ emptiness}}\,. \tag{4.28}$$

This is just the consequence of $a \equiv !a$ we have in equation 2.28 in chapter 2. The negation of 4.28 is just

$$\boxed{\text{emptiness } \equiv \text{ form}}\,, \tag{4.29}$$

which is the consequence of $!a \equiv a$ we have in equation 2.29 in chapter 2. Therefore together we have

$$\text{form } \equiv \text{ emptiness} \quad \text{and} \quad \text{emptiness } \equiv \text{ form} \tag{4.30}$$

Also, form means the presence of phenomenon which is also known as conventional designation, and emptiness means unreal (virtuality). Therefore we can write

$$\boxed{\text{phenomenon } \equiv \text{virtuality}}\,. \tag{4.31}$$

This is the famous statement in the Diamond sutra of every phenomenon (conventional destination) is virtual, and the equivalent statement of any connected dharma is like illusionary dreams and shadows. Furthermore, phenomenon is just illusive being ($!|1\rangle$) in figure 4.2, we have

$$\text{pheonmenon} = !|1\rangle\,, \tag{4.32}$$

therefore,

$$\text{illusive being} \equiv \text{virtuality}\,. \tag{4.33}$$

And as $(!|1\rangle) \equiv \$|0\rangle)$ by 4.1, where $!|0\rangle$ is actual nothing, we can also write

$$\text{actual nothing} \equiv \text{virtuality}\,. \tag{4.34}$$

Thus our equations are consistent in logic. Finally, as mind has creation and annihilation, we infer creation and annihilation as our mind,

$$\text{presence of mind} = \text{creation and annihilation}\,. \tag{4.35}$$

Thus,

$$\boxed{\text{presence of mind} = \text{emptiness} = \text{virtual}}\,. \tag{4.36}$$

The creation and annihilation of mind give rise to phenomenon, which is virtual.

Now consider its opposite counterpart,

$$(U^*|S_k) \equiv (U|S_k^\star)\,. \tag{4.37}$$

This means

$$(0|\text{interconnectedness}) \equiv (1|\text{self})\,. \tag{4.38}$$

The term (0|interconnectedness) means that there is an absence of interconnectedness, which everything is not depending on each other. This is referred as unconnected dharma, which is non-phenomenological, it is non-form. The term (1|self) means there is a presence of independent self, termed as being. Such presence is also referred as reality. Mathematically, take

$$\bar{a} = \text{non-form} = (0|\text{interconnectedness}), \tag{4.39}$$

then we have

$$\boxed{\text{non-form} \equiv \text{being}}\,. \tag{4.40}$$

This is just the consequence of $\bar{a} \equiv !\bar{a}$. The negation of 4.40 is

$$\boxed{\text{being} \equiv \text{non-form}}\,. \tag{4.41}$$

This is just the consequence of $!\bar{a} \equiv \bar{a}$. Therefore together we have

$$\text{non-form} \equiv \text{being} \quad \text{and} \quad \text{being} \equiv \text{non-form}\,. \tag{4.42}$$

As non-form means the absence of phenomenon (non-phenomenon), and the presence of independent self is referred as reality. Therefore we have

$$\boxed{\text{non-phenomenon} \equiv \text{reality}}\,. \tag{4.43}$$

Non-creation and non-annihilation of mind means that no phenomenon is formed, we thus have

$$\text{absence of mind} = \text{non-creation and non-annihilation}\,. \tag{4.44}$$

This implies that

$$\boxed{\text{absence of mind} \equiv \text{reality}}\,. \tag{4.45}$$

With all these results, we can write down the full dual equation in different forms.

$$\begin{cases} \text{form} \equiv \text{emptiness} \quad \text{and} \quad \text{emptiness} \equiv \text{form} \qquad \text{for partition } D \\ \text{non-form} \equiv \text{being} \quad \text{and} \quad \text{being} \equiv \text{non-form} \quad \text{for partition } D^* \end{cases}. \tag{4.46}$$

Alternatively,

$$\begin{cases} \text{phenomenon } \equiv \text{virtuality} & \text{for partition } D \\ \text{non-phenomenon} \equiv \text{reality} & \text{for partition } D^* \end{cases} . \tag{4.47}$$

In terms of 4-tableau diagram, we have

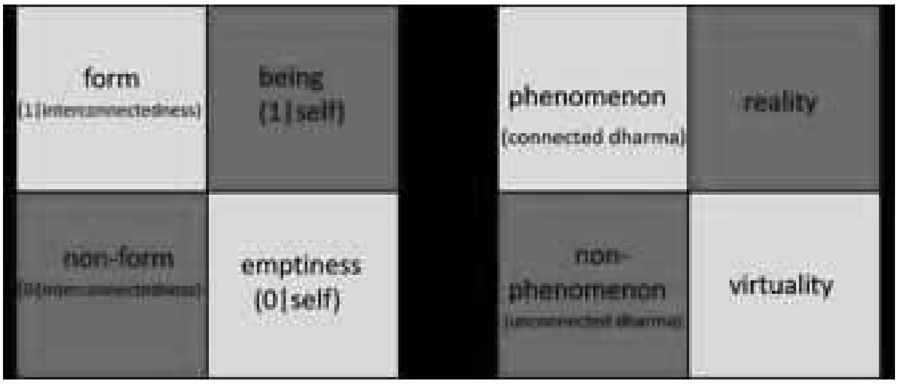


Figure 4.6

Therefore the prajnaparamita is the trivialization of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry.

### 4.1.3 The Middle Way

The thought of middle way is the core part of the Madhyamaka school of Buddhism. The middle way means to remove the two levels of dichotomization, and exclude the middle. According to Mahaprajnaparamitasastra (known as The Treatise on the Great Perfection of Wisdom) volume 6, which is also written by Nagarjuna, says that the middle way is termed as neither existence nor nothing, nor neither existing nor nothing, nor this descriptive statement [23]. According to Mahaprajnaparamitasastra Volume 43, Prajnaparamita is referred as proceeding the middle way without falling into dualities as a conclusive statement [23]. The dualities include different kinds of antagonistic levels, for instance permanence and impermanence, visible and invisible, etc which are described by the dual set $\{u, u^*\}$. In the beginning of the Madhyamaka sutra, it expounds the concept of eight negations: nor created and nor annihilated, nor permanent and nor impermanent, nor same nor different, nor to and nor fro.

To demonstrate the concept of middle way, we will use the mathematics of 4-box tableaux we constructed in chapter 3. Let $U = \{u\}$ and $U^* = \{u^*\}$. We have the following

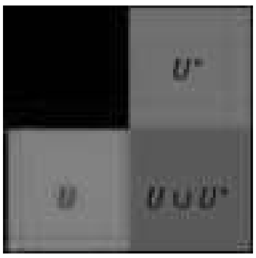


Figure 4.7

The fullness is considered as $U \cup U^*$. By taking the negation,

$$*(U \cup U^*) = *U \cap *U^* = *U \cap U = U^* \cap U = \emptyset\,, \tag{4.48}$$

where in the second step we have used the de-Morgan's theorem. For example, let $u =$ creation, $u^* =$ annihilation, the fullness is creation or annihilation. Not creation or annihilation is nor creation and nor annihilation, which we will prove here is indeed the empty set. Here we have also excluded the middle of $\{0\}$ as we do not include it in our definition, so the exclusion of middle is automatically satisfied. In this session We will prove that the middle way infers the empty set $\emptyset$, while its dual infers the full set.

First we would like investigate some their properties. We have respectively the full set and the null set as

$$W = U \cup U^* \quad \text{and} \quad \emptyset = U \cap U^*\,. \tag{4.49}$$

Explicitly,

$$W = \{u, u^*\} \quad \text{and} \quad \emptyset = \{\}\,. \tag{4.50}$$

The respective cardinality is $|W| = 2$ and $|\emptyset| = 0$. The full set and the empty set are dual to each other,

$$\emptyset = *W \quad \text{and} \quad W = *\emptyset\,, \tag{4.51}$$

which is shown by the de-Morgan's law. In a detailed look, this can be manifested as

$$(1)\emptyset = (*)W \quad \text{and} \quad (1)W = (*)\emptyset\,, \tag{4.52}$$

where 1 is the identity operator element while $*$ is the dual operator element. We can take $u = 1$ and $u^* = *$ and the abstract perspective $S_k = W$ and $S_k^\star = \emptyset$, then 4.51 means to be the standard form,

$$(1|\emptyset) \equiv (*|W) \quad \text{and} \quad (1|W) \equiv (*|\emptyset)\,. \tag{4.53}$$

In a compact form, this is just the standard relation we had before,

$$(1|0) \equiv (0|1) \quad \text{and} \quad (1|1) \equiv (0|0)\,. \tag{4.54}$$

Similarly, for the half sets,

$$U = \{u\} \quad \text{and} \quad U^* = \{u^*\}\,. \tag{4.55}$$

The two half sets are dual to each other,

$$U^* = *U \quad \text{and} \quad U = *U^*\,. \tag{4.56}$$

Explicitly this means

$$(1|U^*) \equiv (*|U) \quad \text{and} \quad (1|U) \equiv (*|U^*)\,. \tag{4.57}$$

and the compact form shares the one in 4.54.

Element-wise, figure 4.7 can be represented as follow,

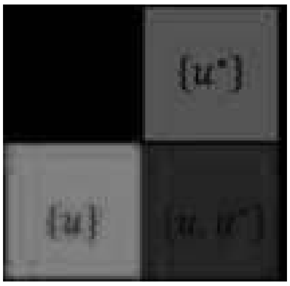


Figure 4.8

**The canonical 4-duality structure revisited**

Now, we will return to the study of the core thought of the Middle way. First consider the first statement of negation: "nor created and nor annihilated". In terms of state, we have $|u\rangle = |\text{created}\rangle$ and its dual state $|u^*\rangle = |\text{annihilated}\rangle$. The negation nor operator is defined as $\hat{P}_1 = \star\hat{e} = \hat{-}$. So the statement of "nor created and nor annihilated" is mathematically represented as

$$\begin{aligned} &\star\, \hat{e}(|\text{created}\rangle \oplus |\text{annihilated}\rangle) \\ &= (\star\hat{e} \oplus \star\hat{e})(|\text{created}\rangle \oplus |\text{annihilated}\rangle) \\ &= \star\hat{e}|\text{created}\rangle \oplus \star\hat{e}|\text{annihilated}\rangle \\ &= \hat{-}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle\,. \end{aligned} \tag{4.58}$$

However, there also exists a state with the dual of it. The normal operator is defined as $\hat{P}_0 = \star\hat{e} \star \hat{e} = \hat{e} = \hat{+}$,

$$\begin{aligned} &\star\, \hat{e}\big(\star\, \hat{e}(|\text{created}\rangle \oplus |\text{annihilated}\rangle)\big) \\ &= (\star\hat{e} \oplus \star\hat{e})(\star\hat{e} \oplus \star\hat{e})(|\text{created}\rangle \oplus |\text{annihilated}\rangle) \\ &= \hat{e}|\text{created}\rangle \oplus \hat{e}|\text{annihilated}\rangle \\ &= \hat{+}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle\,. \end{aligned} \tag{4.59}$$

This is the statement of "being created and being annihilated", which is the negation statement of "nor created and nor annihilated".

Finally we have two more states, "being created and nor annihilated" and its negation "nor created and being annihilated", which are mathematically, respectively,

$$\hat{+}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle \quad \text{and} \quad \hat{-}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle \tag{4.60}$$

Therefore, the full state consists of two dual pairs as follow,

$$\begin{aligned} \psi = &(\hat{+}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle) \oplus (\hat{-}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle) \\ &\oplus (\hat{+}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle) \oplus (\hat{-}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle) \end{aligned} \tag{4.61}$$

Or neatly we write symbolically as a canonical 4-duality structure,

$$\psi = (\hat{+}|u\rangle \oplus \hat{+}|u^*\rangle) \oplus (\hat{-}|u\rangle \oplus \hat{-}|u^*\rangle) \oplus (\hat{+}|u\rangle \oplus \hat{-}|u^*\rangle) \oplus (\hat{-}|u\rangle \oplus \hat{+}|u^*\rangle), \tag{4.62}$$

which is just 4.21. In terms of 4-tableau, we have

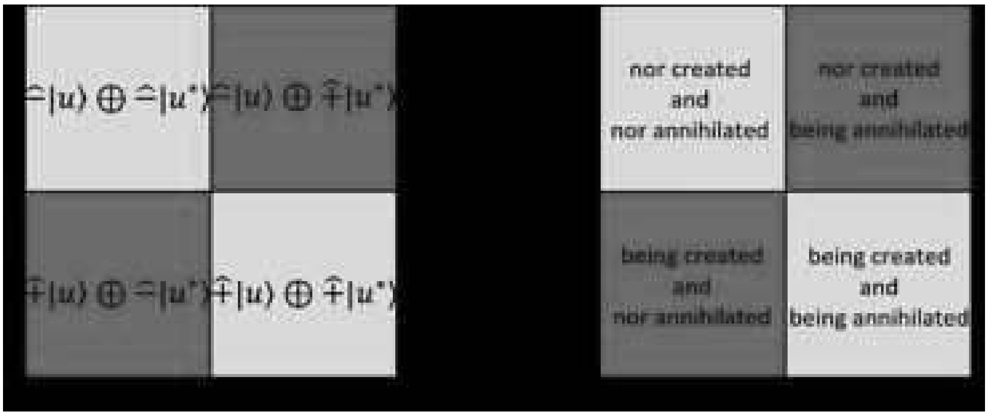


Figure 4.9

Therefore, the middle way is the illustration of the $\hat{-}|u\rangle \oplus \hat{-}|u^*\rangle$ state, which is one of the four states of the canonical structure. On the other hand, its dual is the illustration of the $\hat{+}|u\rangle \oplus \hat{+}|u^*\rangle$ state. Now, since ($\hat{+}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle$) means that appearance is created and annihilated, this is regarded as the connected dharma, this is the view of the ordinary sentient being. Then ($\hat{-}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle$) is that state of no appearance, which also infers unconnected dharma, this is the view of the Buddha, and so does the middle way. To sum up,

$$\hat{+}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle \implies \text{presence of appearance} \implies \text{sentient being's view} \tag{4.63}$$

and

$$\hat{-}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle \implies \text{absence of appearance} \implies \text{Buddha's view}\,. \tag{4.64}$$

We remain to have two duals states left, these are the half states. They do not refer to either the view of ordinary sentient beings or the view of the Buddha.

Now, we evaluate the operations. First we can see that

$$\hat{+}|i\rangle = |i\rangle \quad \text{and} \quad \hat{-}|i\rangle = \bar{\mathbf{0}}_i = \bar{\mathbf{0}}\,. \tag{4.65}$$

Here, $\bar{\mathbf{0}}$ is defined as a empty vector which does not contain any entries, it is a hollow column vector with zero dimension, which we call the vacuum state,

$$\bar{\mathbf{0}} = \begin{pmatrix} \quad \end{pmatrix}, \tag{4.66}$$

so that for all $|v\rangle \in V$, $\bar{\mathbf{0}} \oplus |v\rangle = |v\rangle \oplus \bar{\mathbf{0}}$ and $\bar{\mathbf{0}} \oplus \bar{\mathbf{0}} = \bar{\mathbf{0}}$.

This is because $\hat{+}$ is the being-operator, this gives the state itself; while $\hat{-}$ is the nor-operator, the action on the state means the state does not exist, so this gives the empty vector. Therefore, 4.62 is evaluated to be

$$\begin{aligned} \psi &= (|u\rangle \oplus |u^*\rangle) \oplus (\bar{\mathbf{0}}_u \oplus \bar{\mathbf{0}}_{u^*}) \oplus (|u\rangle \oplus \bar{\mathbf{0}}_{u^*}) \oplus (\bar{\mathbf{0}}_u \oplus |u^*\rangle) \\ &= (|u\rangle \oplus |u^*\rangle) \oplus \bar{\mathbf{0}} \oplus |u\rangle \oplus |u^*\rangle)\,. \end{aligned} \tag{4.67}$$

This gives the result of figure 3.4. Therefore in terms of direct sum vector space representation, this is figure 3.4,

Figure 4.10

Therefore we have the following results

$$
\begin{aligned}
&\hat{-}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle = \bar{\mathbf{0}}\,. \\
&\hat{+}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle = |\text{created}\rangle \oplus |\text{annihilated}\rangle\,. \\
&\hat{+}|\text{created}\rangle \oplus \hat{-}|\text{annihilated}\rangle = |\text{created}\rangle\,. \\
&\hat{-}|\text{created}\rangle \oplus \hat{+}|\text{annihilated}\rangle = |\text{annihilated}\rangle\,.
\end{aligned}
\tag{4.68}
$$

Explicitly in terms of diagram,

Figure 4.11

In terms of set, we have 4.8. We find the structure of the set is identical to the case of direct-sum space. Therefore one has the isomorphism,

Figure 4.12

The operation isomorphism for the two cases hold as follow,

$$\begin{aligned} U \cup \emptyset = \emptyset \cup U = U \quad &, \quad |v\rangle \oplus \bar{\mathbf{0}} = \bar{\mathbf{0}} \oplus |v\rangle = |v\rangle \,. \\ U^* \cup \emptyset = \emptyset \cup U^* = U^* \quad &, \quad |v^*\rangle \oplus \bar{\mathbf{0}} = \bar{\mathbf{0}} \oplus |v^*\rangle = |v^*\rangle \,. \end{aligned} \tag{4.69}$$

In addition, the power set of the full set $W$ is the discrete topology of $W$,

$$\tau = \{\emptyset, \{u\}, \{u^*\}, \{u, u^*\}\} \,. \tag{4.70}$$

This is isomorphic to the binary number set of

$$\tau = \{00, 10, 01, 11\} \,. \tag{4.71}$$

In conclusion, the middle-way, which is "nor created and nor annihilated" means empty space (set) mathematically. We find that the empty space (set) is the correct description of the middle way. In the scholastic description of the middle way, it is stated that neither $u$ (one side), nor $u^*$ (the other dual side) , and not the middle (0). In our mathematical description, 0 is not the middle way, while nothingness is. The Buddhist view infers the reality has no appearance, non-phenomenological. Since the empty set does not contain any elements, this implies such object has no appearance. Thus it is a good description of the Buddhist view, the middle way. The dual perspective is the view the ordinary sentient beings, which is the full set (space). It contains both the appearance of creation and the appearance of annihilation.

### Cardinality representation

Finally, we would like to work on the cardinality representation of the above representation. If there exists a $u$ or $|u^\star\rangle$, we label as 1; while if it is $\mathbf{0}$ then we have 0. In other words we get,

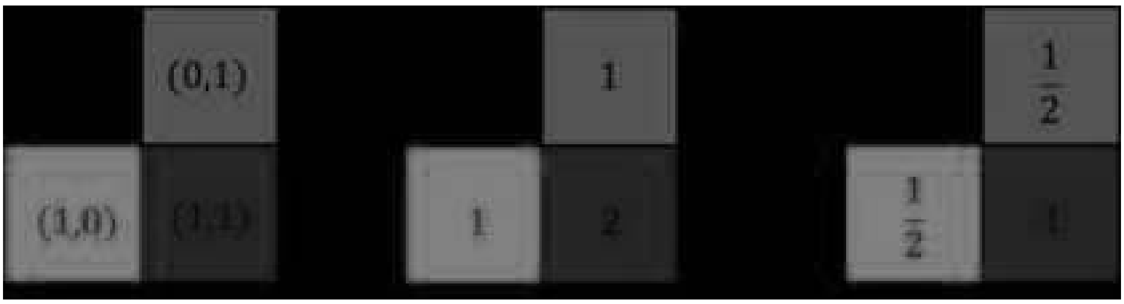


Figure 4.13

In the first diagram, we can actually see that the cardinality representation is the representation of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ group. In the right diagram, we calculate the cardinality of the set for each quadrant. In the last diagram, the 1/2 representation denotes the red and blue quadrant are half quadrants , while the 0 representation is the empty quadrant and the 1 representation is the full quadrant. Next, we would like to construct the cardinality representation matrix by directing summing the cardinality elements of the state,

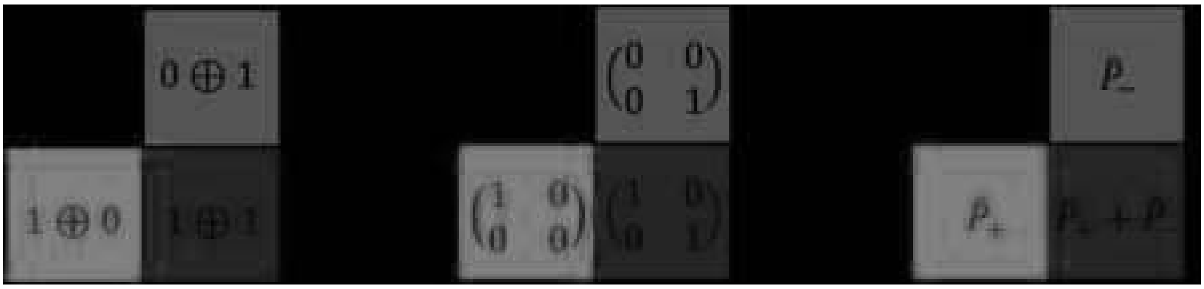


Figure 4.14

Now define the dual doublet as the full state

$$\phi = \begin{pmatrix} |u\rangle \\ |u^*\rangle \end{pmatrix} = |u\rangle \oplus |u^*\rangle \,. \tag{4.72}$$

Each representation matrix can be defined by

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad , \quad \mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad , \quad \hat{P}_+ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad , \quad \hat{P}_- = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \,. \tag{4.73}$$

where $\hat{P}_+$ and $\hat{P}_-$ are projection matrices constructed by the elements of the cardinality, which satisfies

$$\hat{P}_+ + \hat{P}_- = \mathbf{1} \quad , \quad \hat{P}_+\hat{P}_- = \hat{P}_-\hat{P}_+ = \mathbf{0} \,. \tag{4.74}$$

Define also

$$\underline{\mathbf{0}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0 \oplus 0 \quad , \quad \phi_+ = \begin{pmatrix} |u\rangle \\ 0 \end{pmatrix} = |u\rangle \oplus 0 \quad , \quad \phi_- = \begin{pmatrix} 0 \\ |u^*\rangle \end{pmatrix} = 0 \oplus |u^*\rangle \,. \tag{4.75}$$

Then we have

$$\mathbf{1}\phi = \phi \quad , \quad \mathbf{0}\phi = \underline{\mathbf{0}} \quad , \quad \hat{P}_+\phi = \phi_+ \quad , \quad \hat{P}_-\phi = \phi_- \,. \tag{4.76}$$

Also we have

$$\hat{P}_+\phi_- = \underline{\mathbf{0}} \quad , \quad \hat{P}_-\phi_+ = \underline{\mathbf{0}} \,. \tag{4.77}$$

Since $*\phi_+ = \phi_-$ and $*\phi_- = \phi_+$ ; $*\phi = \underline{\mathbf{0}}$ and $*\underline{\mathbf{0}} = \phi$, it follows from equation 4.76 that for operators, we have

$$*\hat{P}_+ = \hat{P}_- \; , \; *\hat{P}_- = \hat{P}_+ \quad ; \quad *\mathbf{1} = \mathbf{0} \; , \; *\mathbf{0} = \mathbf{1} \,. \tag{4.78}$$

We term the second dual equivalence as *associated duality* given in reference [1] section 2.1, for which such duality is not given by normal matrix operation that we introduced before. In addition, for $\hat{P}_+$ and $\hat{P}_-$, the dual operator can be regarded as off-diagonal transpose $S$, such that $\hat{P}_+ = (\hat{P}_+)^S$. We can also check that the dual operators are orthogonal.

$$\hat{P}_+ \cdot \hat{P}_- = \mathrm{Tr}(\hat{P}_+^T \hat{P}_-) = 0 \quad , \quad \hat{P}_- \cdot \hat{P}_+ = \mathrm{Tr}(\hat{P}_-^T \hat{P}_+) = 0 \,, \tag{4.79}$$

and

$$\mathbf{0} \cdot \mathbf{1} = \mathrm{Tr}(\mathbf{0}^T \mathbf{1}) = 0 \quad , \quad \mathbf{1} \cdot \mathbf{0} = \mathrm{Tr}(\mathbf{1}^T \mathbf{0}) = 0 \,. \tag{4.80}$$

The cardinality is given by the norm of the state vector,

$$\begin{aligned}
\mathrm{card}(\underline{\mathbf{0}}) &= \langle\phi|\phi\rangle = \mathrm{Tr}(0\oplus 0) = 0\,,\\
\mathrm{card}(\phi_+) &= \langle\phi_+|\phi_+\rangle = \mathrm{Tr}(1\oplus 0) = 1\,,\\
\mathrm{card}(\phi_-) &= \langle\phi_-|\phi_-\rangle = \mathrm{Tr}(0\oplus 1) = 1\,,\\
\mathrm{card}(\phi) &= \langle\phi|\phi\rangle = \mathrm{Tr}(1\oplus 1) = 2\,.
\end{aligned} \tag{4.81}$$

There is also one important property, note that

$$(\hat{P}_+ + \hat{P}_-)^2 = \hat{P}_+^2 + \hat{P}_-^2\,, \tag{4.82}$$

which takes the form of $(a+b)^2 = a^2+b^2$, this is prohibited when $a, b$ are real numbers (except zero), but allowed when $a, b$ are matrices. Similarly, we have

$$(\hat{P}_+ + \hat{P}_-)^2 = (\hat{P}_+ - \hat{P}_-)^2 \tag{4.83}$$

and this only holds for the matrix case..

### 4.1.4 The general case

Now we would like to work out for the general case when there is more than one creation and annihilation. For convenience, here we will use the set to demonstrate the idea instead of vector space, but it should be noted that the idea would be similar for the latter case. Let $U_i = \{u_i\}$ and its dual $U_i = \{u_i^*\}$, define

$$\mathcal{U} = U_1 \cup U_2 \cup \cdots \cup U_n = \bigcup_{i=1}^{n} U_i = \{u_1, u_2, \cdots, u_n\}\,, \tag{4.84}$$

and its dual

$$\mathcal{U}^* = U_1^* \cup U_2^* \cup \cdots \cup U_n^* = \bigcup_{i=1}^{n} U_i^* = \{u_1^*, u_2^*, \cdots, u_n^*\}\,, \tag{4.85}$$

The full set All is given by the union of $\mathcal{U}$ and $\mathcal{U}^*$,

$$\begin{aligned}
\mathcal{W} &= \mathcal{U} \cup \mathcal{U}^*\\
&= (U_1 \cup U_2 \cup \cdots \cup U_n) \cup (U_1^* \cup U_2^* \cup \cdots \cup U_n^*)\\
&= (U_1 \cup U_1^*) \cup (U_2 \cup U_2^*) \cup \cdots \cup (U_n \cup U_n^*)\\
&= \bigcup_{i=1}^{n} (U_i \cup U_i^*)\\
&= W_1 \cup W_2 \cup \cdots \cup W_n\,.
\end{aligned} \tag{4.86}$$

Hence the full set is just $\mathcal{W} = \{u_1, u_1^*; u_2, u_2^*; \cdots; u_n, u_n^*\}$. Next we would like to find out the intersection for the negation, in fact we will show that it is an empty set as expected.

$$\begin{aligned}
*\mathcal{W} &= *W_1 \cap *W_2 \cap \cdots \cap *W_n\\
&= \emptyset \cap \emptyset \cap \cdots \cap \emptyset\\
&= \emptyset\,.
\end{aligned} \tag{4.87}$$

Also we have $\mathcal{W} = *\emptyset$. Diagramatically we have

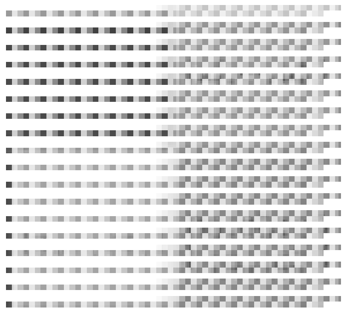

Figure 4.15

### 4.1.5 The Heart Sutra, Middle way and Prajnaparamita

In the last section, we will reconnect the core concept in Middle way and Prajnaparamita in the Heart Sutra. First, one notices that "being" is presence of interconnection is (1|interconnection), which means the presence of creation and annihilation of thoughts. This is because appearance can be created and destroyed. We have

$$(1|\text{interconnection}) = \text{creation and annihilation}\,. \tag{4.88}$$

It follows that

$$\text{depending arising} = \text{form} = \text{creation and annihilation} \equiv \text{emptiness}\,. \tag{4.89}$$

And since creation and annihilation infer impermanence, this implies

$$\boxed{\text{impermanence} \equiv \text{emptiness}}\,. \tag{4.90}$$

Next we also have

$$\text{phenomenon} = \text{creation and annihilation} \tag{4.91}$$

Finally by equations 4.31 and 4.91, we arrive at the conclusion that

$$\text{creation and annihilation} \equiv \text{virtuality}\ \ . \tag{4.92}$$

and

$$\boxed{\text{All} \equiv \text{virtuality}\ }\,, \tag{4.93}$$

where we recall the set "All" means the union of all creations and annihilations. That means for phenomenon that can be created and destroyed, they are intrinsically virtual. This result is profound, as it relates the the dual thought of the middle way to the Diamond Sutra.

On the other hand, for the dual part,

$$(0|\text{interconnection}) = \text{non-creation and non-annihilation}\,. \tag{4.94}$$

It follows that

$$\text{non-depending arising} = \text{non-form} = \text{non-creation and non-annihilation} \equiv \text{being}\,. \tag{4.95}$$

Therefore we also have

$$\text{non-phenomenon} = \text{non-creation and non-annihilation} \ . \tag{4.96}$$

By equations 4.43 and 4.96, we finally arrive at the conclusion of

$$\text{non-creation and non-annihilation} \ \equiv \text{reality}. \tag{4.97}$$

Since recalling that (non-creation and non-annihilation ) $= \emptyset$, we have

$$\boxed{\emptyset \equiv \text{reality}}\,. \tag{4.98}$$

$\emptyset$ refers to nothingness. So the reality is nothingness. In terms of direct-sum vector space, this is the vacuum state $|\bar{\mathbf{0}}\rangle$. Therefore the reality is given by the vacuum. Finally non-creation and non-annihilation infer permanence, so we have

$$\boxed{\text{permanance} \equiv \text{reality}}\,. \tag{4.99}$$

Our proof clarifies the distinction between emptiness and nothingness, for which they are two entirely different concepts and not equal to each other. We can see that for emptiness,

$$\text{emptiness} \equiv \text{creation and annihilation}\,. \tag{4.100}$$

which is the absence of an independent-self; while for nothingness,

$$\text{nothingess} = \emptyset = \text{non-creation and non-annihilation} \equiv \text{being}\,. \tag{4.101}$$

Therefore the concept of emptiness and nothingness is dual to each other. In other words,

$$\boxed{*\text{empintess} = \text{nothingness} \quad , \quad *\text{nothingness} = \text{empintess}}\,. \tag{4.102}$$

In Mūlamadhyamakakārikā (The treatise of the middle) composed by Nagarjuna, the concept of two satya is introduced: conventional truth and provisional truth. It is written as follows in the original text: "The conventional truth is existence with conventional designation, the provisional truth is ultimate emptiness " [24]. The existence here does not mean actual existence, but illusionary existence because existence is formed by causes and conditions (dependent arising), for which is ultimately empty ontologically. It also says "The dharma which is dependent arising, I say it is empty, also termed as conventional designation, which is the middle way. There is no dharma which is not formed by causes and conditions, thus all dharma is empty." This is referred as "All" in the left hand side of equation 4.93, which is the conventional truth. "All" is empty, and virtual, thus this is the right hand side of equation 4.93. Mathematically, we can write

$$(1|\text{conventional truth}) \equiv (0|\text{provisional truth})\,. \tag{4.103}$$

The is interpreted as, existence in conventional truth is equivalent to emptiness in provisional truth. Therefore, conventional truth and provisional truth is dual to each other, as expected. Therefore, the conventional truth cannot be held without the provisional truth, and the provisional truth cannon be held without the conventional truth.

So far we have seen dualities in All and $\emptyset$, virtuality and duality. However, in the theory of Middle way, it is clearly said that the ultimate truth is not in duality, nor the middle. Since All denotes the being of existence, $\emptyset$ denotes nothingness, so the ultimate truth is neither existence nor nothingness, for which only such negation can fulfil the definition of the Middle way. Mathematically this is $\mathcal{W} \cap \emptyset$, i.e. the intersection of All and nothingness,

$$\mathcal{W} \cap \emptyset = \text{All } \cap \text{nothingness } = \text{nothingness}\,, \tag{4.104}$$

which is still, nothingness. Therefore, mathematically we show that the negation of All and nothingness is still nothingness and this is the reality per se.

## 4.2 Lineage of Madhyamaka school of Buddhism and their major thoughts

The lineage of the Madhyamaka school of Buddhism begins by Nāgārjuna (CE 150-250), followed by Āryadeva (CE 170-270) and Rāhulabhadra (CE 200-300). Later the school divides into two major sects, the svātantrika and prāsaṅgika. Representative figures of svātantrika includes Bhāvaviveka (CE 490-570), followed by Avalokitavrata (CE -700-), Śrīgupta and Jñānagarbha (CE 700-760). Later the sect of svātantrika transforms into the sect of Madhyamaka-Yogācāra, representative figures include Śāntirakṣita (CE 725-790), Kamalaśīla, Vimuktisena and Haribhadra (CE -800-). For the sect of prāsaṅgika, representative figures include Buddhapālita (CE 470-540), followed by Candrakīrti (CE 600-650), Śāntideva (CE 650-700), Prajñākaramati (CE 950-1030) and Atīśa (982-1054).

In particular, from the historical note, the two most famous figures with their canons that arise attention and fruitful research are Bhāvaviveka and Candrakīrti. There are also differences in the views within Madhyamaka Buddhism among the saints mentioned above.

There is a great debate between Bhāvaviveka and Candrakīrti. Although Bhāvaviveka, Candrakīrti, Buddhapālita, etc admit that there exists an realistic external appearance beyond the mind in which the Yogācāra school does not agree, Bhāvaviveka admits that external appearance has independent-self nature, i.e. the conventional truth has a self-independent nature; while Candrakīrti does not admit the existence of independent-self nature of external appearance.

We argue that the claim made by Bhāvaviveka is incorrect while the view of Candrakīrti is correct. This is because if, according to Bhāvaviveka's claim, this will break the equality of equation 4.103. In Bhāvaviveka's claim, we will have (1|conventional truth) and (1|provisional truth), for which they cannot be equal. This breaks the duality equivalence relation. The equality must hold only for (0|provisional truth). In addition, there is a famous example that external appearance depends of mind and karma. According to the Exposition of Mahāyāna-samgraha, due to the difference of karma, the heavenly people perceive water as basement adorned with precious jewelleries; human beings perceive water as clear or contaminated liquid; hungry ghosts observe water as pus and blood; fish observe water as home for survival [27]. Therefore, the external appearance is formed by causes and conditions and it is a reflection of karma, which are all dependent arising, of which themselves must be empty.

## 4.3 Introduction to Tathāgatagarbha's thought

In this section, we will introduce the Tathāgatagarbha's thought in Mahayana Buddhism. Tathāgatagarbha is interpreted as two words, Tathāgata-garbha, where the former word is the other name of the Buddha meaning nor coming and nor fro, while the latter word means storage or matrix. It homes the seed of developing appearances of karma. Tathāgatagarbha's refers to Buddhahood that all the sentient being originally possess and ultimate reality. It is nor created and nor annihilated, and constant. In other words, this entails Tathāgatagarbha is the empty set, or the vacuum. We have

$$\boxed{\text{Tathāgatagarbha} = \emptyset}\,. \tag{4.105}$$

Following equation 4.98, this leads to

$$\boxed{\text{Tathāgatagarbha} \equiv \text{reality}}\,. \tag{4.106}$$

Formally Tathāgatagarbha has three classifications according to the Discourse of Queen Śrīmālā (Śrīmālādevī-sūtra).

1. Empty Tathāgatagarbha: It has no appearance, nor created and nor destroyed and has a constant, permanent ontology. It is also called thusness (or suchness) (tathatā) and belongs to the unconnected dharma.

2. Non-empty Tathāgatagarbha: It creates phenomenon in the perspective of dependent arising and it belongs to connected dharma.

3. Empty and non-empty Tathāgatagarbha: It is the combination of empty Tathāgatagarbha and non-empty Tathāgatagarbha .

According to the doctrine of Awakening of Faith in the Mahāyāna (Mahāyāna śraddhotpāda śāstra), the original single mind is dichotomized into two divisions [26]:

1. The door of thusness

2. The door of creation and annihilation

Since thusness refers to non-creation and non-annihilation, the two doors are dual to each other. It can be seen that the two doors advocated in Mahāyāna śraddhotpāda śāstra is identical to the first two classifications of Tathāgatagarbha, this implies

$$\text{empty Tathāgatagarbha} = \text{the door of thusness} \tag{4.107}$$

and

$$\text{non-empty Tathāgatagarbha} = \text{the door of creation and annihilation}\,. \tag{4.108}$$

This duality can be equated by introducing a dual perspective

$$(\hat{+}|u\rangle \oplus \hat{+}|u^*\rangle|S_k) \equiv (\hat{-}|u\rangle \oplus \hat{-}|u^*\rangle|S_k^\star)\,, \tag{4.109}$$

and

$$(\hat{-}|u\rangle \oplus \hat{-}|u^*\rangle|S_k) \equiv (\hat{+}|u\rangle \oplus \hat{+}|u^*\rangle|S_k^\star)\,. \tag{4.110}$$

We can take $S_k$ = sentient being and $S_k^\star$ = Buddha. Or in terms of set

$$(*\emptyset|\text{sentient being}) \equiv (\emptyset|\text{Buddha})\,. \tag{4.111}$$

and

$$(\emptyset|\text{sentient being}) \equiv (*\emptyset|\text{Buddha})\,. \tag{4.112}$$

Only the first equation 4.111 is relevant, while 4.112 is meaningless and can be neglected.

Finally for empty and non-empty Tathāgatagarbha, it can be regarded as the admission of empty Tathāgatagarbha and non-empty Tathāgatagarbha, and the negation of them. First let us consider the case of negation. For example, in the Diamond sutra, the Buddha says a statement like "not connected dharma and not unconnected dharma". This implies "not dependent arising and not non-dependent arising". Or explicitly, "nor created and destroyed, and nor not created and destroyed". Mathematically this means

$$*\emptyset \cap \emptyset = \emptyset\,. \tag{4.113}$$

Therefore this is still, the empty set which is nothingness and hence this is the Buddhist view. In the Avatamsaka sutra, the Buddha says nor birth and death, and nor nirvana. Birth and death is the state of dependent arising which is created and destroyed, and nirvana is the state of non-created and non-destroyed. Therefore "nor birth and death, and nor nirvana" still means nothingness, thus this is still retained as nirvana itself.

On the dual side, the admission of the statement is "being created and destroyed, and being not created and destroyed". This is thus

$$*\emptyset \cup \emptyset = *\emptyset\,. \tag{4.114}$$

This is just the full set, and thus the sentient being's view.

## 4.4 Ālaya-vijñāna

Ālaya-vijñāna, also referred as sarva-bījaka-vijñāna (the sense of all seeds) , vipākavijñāna (the sense of different fruition). It is commonly known as asta-vijñāna, which is called the eighth sense. The concept of ālaya-vijñāna originally comes from the Mahāyānābhidharma sūtra [28]. It is advocated by Yogācāra school of Buddhism based on the samdhi-nirmocana-sūtra. The Ālaya-vijñāna is the storage of seeds of vijñāna, causation and fruition, which gives rise to all appearances of karma [28]. Ālaya-vijñāna associates with incessant rising conditions and cognition, its contaminated part plays a part with kleshas which addresses sakkāya-diṭṭhi (the view of body and aggregates), arrogance, self-attachment and ignorance [28]. Unenlightened sentient beings are amused with ālaya-vijñāna and do not recognise that it is the foundation of incessant birth and death [28].

Now we will investigation the relationship between the Ālaya-vijñāna and Tathāgatagarbha. According to the doctrine of Awakening of Faith in the Mahāyāna, it expounds that the mind with creation and annihilation associates with the Tathāgatagarbha. The ālaya-vijñāna is the combination of non-creation, non-annihilation and creation, annihilation. Tathāgatagarbha and ālaya-vijñāna is neither same nor different. Mathematically, the combination means the union of two sets here. Since non-creation and non-annihilation is thusness, which is $\emptyset$ ; while its dual, creation and annihilation is the full set $\mathcal{W} = *\emptyset$, their combination is just

$$\mathcal{W} \cup \emptyset = \mathcal{W} . \tag{4.115}$$

Therefore, the ālaya-vijñāna is the full set,

$$\boxed{\text{ālaya-vijñāna} = \mathcal{W} \cup \emptyset = \mathcal{W} = *\emptyset} . \tag{4.116}$$

In line with the doctrine of Awakening of Faith in the Mahāyāna in the last section, we can see that the full set $\mathcal{W} = *\emptyset$ is one mind, which opens up two doors, the door of creation and annihilation $\{u, u^*\}$, and its dual the door of thusness (nor-created and nor-annihilated) $\emptyset$.

$$\{u_1, u_1^*; \cdots ; u_n, u_n^*\} = \{u_1, u_1^*; \cdots ; u_n, u_n^*\} \cup \emptyset . \tag{4.117}$$

In order to understand how they are not same nor different, observing that the full set is not the same as the empty set, implying the ālaya-vijñāna $\mathcal{W}$ is not same as Tathāgatagarbha $\emptyset$, in which ālaya-vijñāna is dual to Tathāgatagarbha.

$$\boxed{\text{ālaya-vijñāna} = *\text{Tathāgatagarbha} \quad , \quad \text{Tathāgatagarbha} = *\text{ālaya-vijñāna}} . \tag{4.118}$$

However, they are not different at the same time because the combination of the two sets ālaya-vijñāna and Tathāgatagarbha is still the same set, Tathāgatagarbha.

The ālaya-vijñāna is not divorced from Tathāgatagarbha. We are originally the Buddha, that means all of us possess the clean, uncontaminated Tathāgatagarbha. However, with a mind of ignorance, such perturbation causes creation and annihilation and cover up the original Buddhahood we have, where we see the full set $\{u_1, u_1^*; \cdots ; u_n, u_n^*\}$ of existence. Yet we do not perceive the Tathāgatagarbha, which is the empty set $\emptyset$.

## 4.5 Introduction to the Yogācāra school of Buddhism

The sect of Yogācāra school is also known as the school of Vijñānavāda (sense only), is the counterpart of school of the Middle way. The doctrine of Yogācāra school is set on the samdhi-nirmocana-sūtra and Yogācārarabhūmi-śāstra.

Fundamentally, the Yogācāra school advocates three concepts of svabhāva, known as tri-svabhāva, which are introduced as follow [28]:

1. Parikalpita-svabhāva : Sentient beings abide in five aggregates, twelve sites and eighteen realms, and wrongly recognises them possessing an independent, permanent-self (svabhāva). They hold onto such illusionary existence.

2. Paratantra-svabhāva : Everything is formed by causes and conditions, which is of the absence of svabhāva. The universe depends on causes and condition and manifests as virtual phenomenon.

3. Parinispanna-svabhāva: The entire detachment of the illusion of parikalpita-svabhāva, and thorough understanding of paratantra-svabhāva, revealing the ultimate truth of the cosmo.

According to the Vijñānavāda school, it reckons that there is no phenomenon beyond vijñāna, the three realms (trai-lokya:kāma-dhātu, rūpa-dhātu and ārūpya-dhātu) is embodied by ālaya-vijñāna. The existence of tri-svabhāva is based on the premise of the existence of ālaya-vijñāna [28]. The Vijñānavāda school admits that ālaya-vijñāna is real. Practitioners of Yogācāra school transform the ālaya-vijñāna into wisdom.

### 4.5.1 Transformation of vijñāna to wisdom

In this section, we will study the concept of Yogācāra school mathematically in details. We will also study how ālaya-vijñāna can be transformed into wisdom provided in the text of Mahāyāna-samgraha. The details of the theory is described in volume 3 [28].

In Mahāyāna-samgraha, both ālaya-vijñāna and non-ālaya-vijñāna are defined. The definition of ālaya-vijñāna is defined above, non-ālaya-vijñāna refers to the dharmakāya of the Buddha which liberation is attained. The dharmakāya is non-created and non-annihilated, which is a constant entity. This distinction of ālaya-vijñāna and non-ālaya-vijñāna is just the two doors introduced in the doctrine of Awakening of Faith in the Mahāyāna, but with different designation. Ālaya-vijñāna is equivalent to the door of creation and annihilation, which is the All set; and non-ālaya-vijñāna is equivalent to the door of thusness, which is non-creation and non-annihilation, which is the empty set.

For convenience take

$$|u\rangle = |0\rangle = |\text{creation}\rangle = \begin{pmatrix}1\\0\end{pmatrix} \quad , \quad |u^*\rangle = |1\rangle = |\text{annihilation}\rangle = \begin{pmatrix}0\\1\end{pmatrix} . \tag{4.119}$$

By isomorphism, we take an ansatz that we can construct the quantum states as follow,

$$\bar{\mathbf{0}} \to |0 \oplus 0\rangle \quad , \quad |u\rangle \oplus \bar{\mathbf{0}} \to |1 \oplus 0\rangle \quad , \quad \bar{\mathbf{0}} \oplus |u^*\rangle \to |0 \oplus 1\rangle \quad , \quad |u\rangle \oplus |u^*\rangle \to |1 \oplus 1\rangle . \tag{4.120}$$

where $|a \oplus b\rangle = |a\rangle \oplus |b\rangle$ for $a, b = 0, 1$. The general form takes the following

$$\hat{P}|0\rangle \oplus \hat{Q}|1\rangle = |a\rangle \oplus |b\rangle , \tag{4.121}$$

where $\hat{P}, \hat{Q}$ are $\hat{+}$ or $\hat{-}$.

In the original representation

$$\hat{-} = \mathbf{0} = \begin{pmatrix}0 & 0\\0 & 0\end{pmatrix} \quad , \quad \hat{+} = \mathbf{1} = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} \tag{4.122}$$

where $\hat{-}$ and $\hat{+}$ are dual to each other as we have stated above. Then we have the following computation,

$$\hat{-}|0\rangle \oplus \hat{-}|1\rangle = \begin{pmatrix}0\\0\\0\\0\end{pmatrix} \quad , \quad \hat{+}|0\rangle \oplus \hat{+}|1\rangle = \begin{pmatrix}0\\1\\1\\0\end{pmatrix} , \tag{4.123}$$

$$\hat{+}|0\rangle \oplus \hat{-}|1\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix} \quad , \quad \hat{-}|0\rangle \oplus \hat{+}|1\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix} , \tag{4.124}$$

We can see that $*(\hat{-}|0\rangle \oplus \hat{-}|1\rangle) = \boldsymbol{M}(\hat{-}|0\rangle \oplus \hat{-}|1\rangle) = \hat{+}|0\rangle \oplus \hat{+}|1\rangle)$. Therefore $\hat{-}|0\rangle \oplus \hat{-}|1\rangle)$ and $\hat{+}|0\rangle \oplus \hat{+}|1\rangle)$ are dual and orthogonal to each other. Also $\hat{-}|0\rangle \oplus \hat{-}|1\rangle$ and $\hat{+}|0\rangle \oplus \hat{+}|1\rangle$ are of associated duality, and they are orthogonal.

Now we would like to use another matrix representation of $\hat{-}$ and $\hat{+}$, so that we can obtain the canonical duality in 4.120. We take

$$\hat{-} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad , \quad \hat{+} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} . \tag{4.125}$$

Clearly that $*\hat{-} = \boldsymbol{M}\hat{+}$ and $*\hat{+} = \boldsymbol{M}\hat{-}$.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} . \tag{4.126}$$

Using this canonical representation,

$$\hat{-}|0\rangle \oplus \hat{-}|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |0\rangle \oplus |0\rangle \quad , \quad \hat{+}|0\rangle \oplus \hat{+}|1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = |1\rangle \oplus |1\rangle \, , \tag{4.127}$$

$$\hat{+}|0\rangle \oplus \hat{-}|1\rangle = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = |0\rangle \oplus |1\rangle \quad , \quad \hat{-}|0\rangle \oplus \hat{+}|1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |1\rangle \oplus |0\rangle \, , \tag{4.128}$$

Now we can see that $*(\hat{-}|0\rangle \oplus \hat{-}|1\rangle) = \boldsymbol{M}(\hat{-}|0\rangle \oplus \hat{-}|1\rangle) = (\hat{+}|0\rangle \oplus \hat{+}|1\rangle)$ is dual and orthogonal to each other $\langle 0 \oplus 0|1 \oplus 1\rangle = \langle 0|1\rangle + \langle 0|1\rangle = 0$, while $\hat{+}|0\rangle \oplus \hat{-}|1\rangle$ and $\hat{-}|0\rangle \oplus \hat{+}|1\rangle$ are associated dual and orthogonal $\langle 0 \oplus 1|1 \oplus 0\rangle = \langle 0|1\rangle + \langle 1|0\rangle = 0$ .

Diagramatically,

Figure 4.16

The states $\{|0\oplus 0\rangle, |0\oplus 1\rangle, |1\oplus 0\rangle, |1\oplus 1\rangle\}$ for the basis of $\mathbb{Z}_2\times\mathbb{Z}_2$. The transformation of each of the basis is given by

Figure 4.17

Explicitly, the all six transformation reads

$$\begin{aligned}
\mathbf{1}\oplus\boldsymbol{M}|0\oplus 0\rangle &= (\mathbf{1}\oplus\boldsymbol{M})(|0\rangle\oplus 0\rangle) = |0\rangle\oplus|1\rangle\,,\\
\boldsymbol{M}\oplus\mathbf{1}|0\oplus 1\rangle &= (\boldsymbol{M}\oplus\mathbf{1})(|0\rangle\oplus 1\rangle) = |1\rangle\oplus|1\rangle\,,\\
\boldsymbol{M}\oplus\boldsymbol{M}|0\oplus 0\rangle &= (\boldsymbol{M}\oplus\boldsymbol{M})(|0\rangle\oplus 0\rangle) = |1\rangle\oplus|1\rangle\,,\\
\boldsymbol{M}\oplus\mathbf{1}|0\oplus 0\rangle &= (\boldsymbol{M}\oplus\mathbf{1})(|0\rangle\oplus 0\rangle) = |1\rangle\oplus|0\rangle\,,\\
\mathbf{1}\oplus\boldsymbol{M}|1\oplus 0\rangle &= (\mathbf{1}\oplus\boldsymbol{M})(|1\rangle\oplus 0\rangle) = |1\rangle\oplus|1\rangle\,,\\
\boldsymbol{M}\oplus\boldsymbol{M}|1\oplus 0\rangle &= (\boldsymbol{M}\oplus\boldsymbol{M})(|1\rangle\oplus 0\rangle) = |0\rangle\oplus|1\rangle\,.
\end{aligned} \tag{4.129}$$

It is noted that

$$\boldsymbol{M}\oplus\boldsymbol{M} = (\mathbf{1}\oplus\boldsymbol{M})(\boldsymbol{M}\oplus\mathbf{1}) = (\boldsymbol{M}\oplus\mathbf{1})(\mathbf{1}\oplus\boldsymbol{M})\,. \tag{4.130}$$

We have the Klein-4 group as

$$\mathbb{Z}_2\times\mathbb{Z}_2 = \{\mathbf{1}\oplus\mathbf{1},\mathbf{1}\oplus\boldsymbol{M},\boldsymbol{M}\oplus\mathbf{1},\boldsymbol{M}\oplus\boldsymbol{M}\}\,. \tag{4.131}$$

The identity operator is given by the completeness relation,

$$\mathbf{1} = |00\rangle\langle 00| + |11\rangle\langle 11| + |01\rangle\langle 01| + |10\rangle\langle 01|\,, \tag{4.132}$$

and the duality operator is given by

$$\boldsymbol{M} = |00\rangle\langle 11| + |11\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01|\,. \tag{4.133}$$

Now we define an qubit operator

$$\hat{\Psi}(\theta) = \cos\theta(\hat{-}\oplus\hat{-}) + \sin\theta(\hat{+}\oplus\hat{+})\,. \tag{4.134}$$

where the phase dependent qubit is

$$|\Psi(\theta)\rangle = \hat{\Psi}(\theta)|u\oplus u^*\rangle\,. \tag{4.135}$$

The qubit operator acts on the full state $|u\oplus u^*\rangle$,

$$\begin{aligned}
\hat{\Psi}(\theta)|u\oplus u^*\rangle &= \cos\theta(\hat{-}|u\rangle\oplus\hat{-}|u^*\rangle) + \sin\theta(\hat{+}|u\rangle\oplus\hat{+}|u^*\rangle)\\
&= \cos\theta(\hat{-}|0\rangle\oplus\hat{-}|1\rangle) + \sin\theta(\hat{+}|0\rangle\oplus\hat{+}|1\rangle)\qquad,\\
&= \cos\theta|0\oplus 0\rangle + \sin\theta|1\oplus 1\rangle
\end{aligned} \tag{4.136}$$

where the phase dependent qubit is

$$|\Psi(\theta)\rangle = \hat{\Psi}(\theta)|u \oplus u^*\rangle\,. \tag{4.137}$$

Now we can refer $|0 \oplus 0\rangle$ state as $|$non-ālaya-vijñāna$\rangle$ since this refers non-creation and non-annihilation; $|1 \oplus 1\rangle$ state as $|$ālaya-vijñāna$\rangle$ since this refers creation and annihilation, where which they are dual to each other. Thus we write

$$|\Psi(\theta)\rangle = \cos\theta|\text{non-ālaya-vijñāna}\rangle + \sin\theta|\text{ālaya-vijñāna}\rangle\,, \tag{4.138}$$

which is equivalent to

$$|\Psi(\theta)\rangle = \cos\theta|\text{nothingness}\rangle + \sin\theta|\text{ālaya-vijñāna}\rangle\,, \tag{4.139}$$

where $|$nothingness$\rangle = *|$ālaya-vijñāna$\rangle$ and $*|$nothingness$\rangle = |$ālaya-vijñāna$\rangle$. According to Mahāyāna-samgraha volume 3, non-ālaya-vijñāna and ālaya-vijñāna transforms. When ālaya-vijñāna is extinguished, non-ālaya-vijñāna arises. This is governed by the phase of the equation. The quantity amount of ālaya-vijñāna and non-ālaya-vijñāna are governed by the probability.

$$P_{\text{non-ālaya-vijñāna}}(\theta) = \langle 0 \oplus 0|\Psi(\theta)\rangle = \cos^2\theta\,, \tag{4.140}$$

and

$$P_{\text{ālaya-vijñāna}}(\theta) = \langle 1 \oplus 1|\Psi(\theta)\rangle = \sin^2\theta\,. \tag{4.141}$$

Therefore, the probability of non-ālaya-vijñāna and ālaya-vijñāna transforms one-another. In particular we have

$$|\Psi(0)\rangle = |\text{non-ālaya-vijñāna}\rangle = |\text{nothingness}\rangle = |\text{Buddha}\rangle\,, \tag{4.142}$$

with probability equal to 1 for the Buddha state. And

$$|\Psi(\pi/2)\rangle = |\text{ālaya-vijñāna}\rangle = |\text{sentient beings}\rangle\,. \tag{4.143}$$

with probability equal to 1 for the sentient being state. The two states are orthogonal,

$$\langle\Psi(0)|\Psi(\pi/2)\rangle = \langle\text{Buddha}|\text{sentient beings}\rangle = 0\,. \tag{4.144}$$

There is a special state when $\theta = \pi/2$, then we have a state which is in analogy to the entangled Bell state (an EPR state),

$$|\Psi(\pi/4)\rangle = \frac{1}{\sqrt{2}}(|0 \oplus 0\rangle + |1 \oplus 1\rangle) \tag{4.145}$$

which is equal to

$$|\Psi(\pi/4)\rangle = \frac{1}{\sqrt{2}}(|\text{nothingness}\rangle + |\text{All}\rangle) \tag{4.146}$$

which can be interpreted as

$$|\Psi(\pi/4)\rangle = \frac{1}{\sqrt{2}}(|\text{reality}\rangle + |\text{virtuality}\rangle) \tag{4.147}$$

or

$$|\Psi(\pi/4)\rangle = \frac{1}{\sqrt{2}}(|\text{Buddha}\rangle + |\text{sentient beings}\rangle)\,. \tag{4.148}$$

or

$$|\Psi(\pi/4)\rangle = \frac{1}{\sqrt{2}}(|\text{unconnected dharma}\rangle + |\text{connected dharma}\rangle)\,. \tag{4.149}$$

This is same as the famous case of Schrodinger's cat in physics, where the cat is regarded as nor alive nor dead, or both alive or dead, unless measurement is taken. Therefore, the ultimate truth here is a qubit, which is is neither unconnected dharma nor connected dharma, or both unconnected dharma and connected dharma, which is indeterministic unless an observation is made, where observation means measurement and the action of the mind. Therefore the reality and virtuality are not distinguished, and the Buddha and sentient beings states are not separated.

According to volume 2 of Mahāyāna-samgraha, depending on paratantra-svabhāva, parikalpita-svabhāva manifests into birth and death (reincarnation), while parinispanna-svabhāva embodies as nirvana, Therefore the Buddha is not abided into birth and death, and nirvana. Mathematically, we write

$$\begin{aligned} |\Psi(\pi/4)\rangle &= |\text{paratantra-svabhāva}\rangle \\ &= \frac{1}{\sqrt{2}}(|\text{parikalpita-svabhāva}\rangle + |\text{parinispanna-svabhāva}\rangle) \\ &= \frac{1}{\sqrt{2}}(|\text{birth and death}\rangle + |\text{nirvana}\rangle) \end{aligned} \tag{4.150}$$

And hence the Buddha is neither in birth and death, nor nirvana, unless He made a choice.

Next we will study some properties of the qubit. Define the orthogonal state by differentiating the state with respect to $\theta$, then we have

$$\langle\Psi(\theta)|\frac{d}{d\theta}|\Psi(\theta)\rangle = 0\,. \tag{4.151}$$

Consider the state where we first take the derivative and the the dual operator,

$$|\psi'(\theta)\rangle = *\frac{d}{d\theta}|\Psi(\theta)\rangle\,. \tag{4.152}$$

This new state will be probability invariant as compared to the original state.

Finally, we are particularly interested in local phase for which $\theta$ is dependent on space and time and is linear of the spacetime variables. In other words, we are interested in the qubit wave operator as

$$\hat{\Psi}(x,y,z,t) = \cos(\omega t - k_x x - k_y y - k_z z)(\hat{+}\oplus\hat{+}) + \sin(\omega t - k_x x - k_y y - k_z z)(\hat{-}\oplus\hat{-}) \tag{4.153}$$

The qubit wave operator acts on the full state $|u \oplus u^*\rangle$,

$$\begin{aligned} \hat{\Psi}|u \oplus u^*\rangle &= \cos(\omega t - k_x x - k_y y - k_z z)(\hat{-}|u\rangle \oplus \hat{-}|u^*\rangle) + \sin(\omega t - k_x x - k_y y - k_z z)(\hat{+}|u\rangle \oplus \hat{+}|u^*\rangle) \\ &= \cos(\omega t - k_x x - k_y y - k_z z)|0 \oplus 0\rangle + \sin(\omega t - k_x x - k_y y - k_z z)|1 \oplus 1\rangle\,. \end{aligned} \tag{4.154}$$

The qubit operator satisfies the wave equation,

$$\Box\hat{\Psi}|u \oplus u^*\rangle = \left(\nabla^2\hat{\Psi} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\hat{\Psi}\right)|u \oplus u^*\rangle = 0 \tag{4.155}$$

where $c$ is the speed of the qubit wave.

Next for the dual qubit wave, we have

$$\hat{\Psi}^{\star}(\theta)=\cos\theta(\hat{+}\oplus\hat{-})+\sin\theta(\hat{-}\oplus\hat{+}) \tag{4.156}$$

By acting on the full state $|u\otimes u^{*}\rangle$,

$$\begin{aligned}\hat{\Psi}^{\star}(\theta)|u\oplus u^{*}\rangle &= \cos\theta(\hat{+}|u\rangle\oplus\hat{-}|u^{*}\rangle)+\sin\theta(\hat{-}|u\rangle\oplus\hat{+}_{3}|u^{*}\rangle)\\ &= \cos\theta(\hat{+}|0\rangle\oplus\hat{-}|1\rangle)+\sin\theta(\hat{-}|0\rangle\oplus\hat{+}|1\rangle) \qquad ,\\ &= \cos\theta|1\oplus 0\rangle+\sin\theta|0\oplus 1\rangle\end{aligned} \tag{4.157}$$

We have

$$\hat{\Psi}^{\star}(\theta)=\cos\theta|\text{creation}\rangle+\sin\theta|\text{annihilation}\rangle \tag{4.158}$$

where $|\text{creation}\rangle=*|\text{annihilation}\rangle$ and $*|\text{creation}\rangle=|\text{annihilation}\rangle$.

$$P_{\text{creation}}(\theta)=\langle 1\oplus 0|\Psi^{\star}(\theta)\rangle=\cos^{2}\theta\,, \tag{4.159}$$

and

$$P_{\text{annihilation}}(\theta)=\langle 0\oplus 1|\Psi^{\star}(\theta)\rangle=\sin^{2}\theta\,. \tag{4.160}$$

In phase equal to zero, we have

$$|\Psi^{\star}(0)\rangle=|\text{creation}\rangle \tag{4.161}$$

with probability equal to 1 for creation state. And

$$|\Psi^{\star}(\pi/2)\rangle=|\text{annihilation}\rangle \tag{4.162}$$

with probability equal to 1 for the annihilation state. The two states are orthogonal,

$$\langle\Psi^{\star}(0)|\Psi^{\star}(\pi/2)\rangle=\langle\text{creation}|\text{annihilation}\rangle=0\,. \tag{4.163}$$

There is a special state when $\theta=\pi/2$, then we have the entangled EPR state as

$$|\Psi(\pi/4)\rangle=\frac{1}{\sqrt{2}}(|1\oplus 0\rangle+|0\oplus 1\rangle) \tag{4.164}$$

which is equal to

$$|\Psi(\pi/4)\rangle=\frac{1}{\sqrt{2}}(|\text{creation}\rangle+|\text{annihilation}\rangle) \tag{4.165}$$

Then the dual wave qubit operator is

$$\hat{\Psi}^{\star}(x,y,z,t)=\cos(\omega t-k_{x}x-k_{y}y-k_{z}z)(\hat{+}\oplus\hat{-})+\sin(\omega t-k_{x}x-k_{y}y-k_{z}z)(\hat{-}\oplus\hat{+})\,. \tag{4.166}$$

By acting on the full state

$$\begin{aligned}\hat{\Psi}^{\star}|u\oplus u^{*}\rangle &= \cos(\omega t-k_{x}x-k_{y}y-k_{z}z)(\hat{+}|u\rangle\oplus\hat{-}|u^{*}\rangle)+\sin(\omega t-k_{x}x-k_{y}y-k_{z}z)(\hat{-}|u\rangle\oplus\hat{+}|u^{*}\rangle)\\ &= \cos(\omega t-k_{x}x-k_{y}y-k_{z}z)|1\oplus 0\rangle+\sin(\omega t-k_{x}x-k_{y}y-k_{z}z)|0\oplus 1\rangle\,.\end{aligned} \tag{4.167}$$

The dual qubit also satisfies the wave equation,

$$\Box\hat{\Psi}^{\star}|u\oplus u^{*}\rangle=\left(\nabla^{2}\hat{\Psi}^{\star}-\frac{1}{c^{2}}\frac{\partial^{2}}{\partial t^{2}}\hat{\Psi}^{\star}\right)|u\oplus u^{*}\rangle=0\,. \tag{4.168}$$

The pair of states are orthogonal to each other

$$\langle \Psi(\theta) | \Psi^\star(\theta) \rangle = 0 \,. \tag{4.169}$$

Note that the linear combination of the two states is also the solution of the wave equation,

$$\phi = A\Psi + B\Psi^\star \,. \tag{4.170}$$

Therefore, when the dual states oscillate, it generates a wave propagation. For example, in equation 4.167, the periodic cycle of creation and annihilation is a wave. Since mind creates and annihilates, this shows that mind is a wave. And for equation 4.154, when the state of nothingness and All oscillate, it creates a wave. So the periodic transformation of ālaya-vijñāna and nothingness will produce a wave signal, that maybe detected in experimental sense.

Next we would like to investigate what happens when the operators act on the dual of $|u \oplus u^*\rangle$, i.e. $*|u \oplus u^*\rangle = |u^* \oplus u\rangle$. Using the representation in 4.125 we find that the following holds,

$$\hat{P} * |0\rangle \oplus \hat{Q} * |1\rangle = \hat{P}|1\rangle \oplus \hat{Q}|0\rangle \,. \tag{4.171}$$

Explicitly, we can check that

$$\begin{aligned}
\hat{-} * |0\rangle \oplus \hat{-} * |1\rangle &= \hat{-}|1\rangle \oplus \hat{-}|0\rangle = |0\rangle \oplus |0\rangle = \hat{-}|0\rangle \oplus \hat{-}|1\rangle \,, \\
\hat{+} * |0\rangle \oplus \hat{+} * |1\rangle &= \hat{+}|1\rangle \oplus \hat{+}|0\rangle = |1\rangle \oplus |1\rangle = \hat{+}|0\rangle \oplus \hat{+}|1\rangle \,, \\
\hat{+} * |0\rangle \oplus \hat{-} * |1\rangle &= \hat{+}|1\rangle \oplus \hat{-}|0\rangle = |0\rangle \oplus |1\rangle = \hat{+}|0\rangle \oplus \hat{-}|1\rangle \,, \\
\hat{-} * |0\rangle \oplus \hat{+} * |1\rangle &= \hat{-}|1\rangle \oplus \hat{+}|0\rangle = |1\rangle \oplus |0\rangle = \hat{-}|0\rangle \oplus \hat{+}|1\rangle \,.
\end{aligned} \tag{4.172}$$

Therefore, we have

$$\hat{-}* = \hat{-} \quad , \quad \hat{+}* = \hat{+} \,. \tag{4.173}$$

We can explicitly check that

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad , \quad \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \,. \tag{4.174}$$

Now we would like to give a more detailed study in the $\boldsymbol{M}$ dual matrix operator. First notice that the action of $* = \boldsymbol{M}$ swap the position of rows as a vertical reflection. For example, for a general matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad \text{then} \quad *A = \boldsymbol{M}A = A^* = \begin{pmatrix} c & d \\ a & b \end{pmatrix} \,. \tag{4.175}$$

If $A$ acts on $\boldsymbol{M}$, the action is to swap the position of the columns as a horizontal reflection,

$$A* = A\boldsymbol{M} = {}^*A = \begin{pmatrix} b & a \\ d & c \end{pmatrix} \,. \tag{4.176}$$

Also we have

$$**A = A** = A \,. \tag{4.177}$$

We have the following identity, recalling $T$ is the transpose action and $R$ is the off-transpose action, we have

$$A^{(TR)} = A^{(RT)} = \boldsymbol{M}A\boldsymbol{M}^{-1} = \boldsymbol{M}A\boldsymbol{M} = *A*^{-1} = *A* \,, \tag{4.178}$$

as $\boldsymbol{M}^{-1} = \boldsymbol{M}$. It follows that $\mathrm{Tr}(A^{(TR)}) = \mathrm{Tr}A$ and $\det(A^{(TR)}) = \det A$. According to equations 4.5.1,

$$*\hat{-}* = \hat{+} \quad , \quad *\hat{+}* = \hat{-}\,. \tag{4.179}$$

And as $* = *^{-1}$, thus $\hat{+}$ and $\hat{-}$ are related by similarity transform

$$*\hat{-}*^{-1} = \hat{+} \quad , \quad *\hat{+}*^{-1} = \hat{-}\,. \tag{4.180}$$

Next we would like to show that the identity

$$(*A*)^{-1} = *A^{-1}* \ . \tag{4.181}$$

This is because $(*A*)^{-1} = *^{-1}(*A)^{-1} = *A^{-1}*$.

Based on the result of 4.172, if follows that $|u^* \oplus u\rangle$ also satisfies the wave equation,

$$\Box\hat{\Psi}|u^* \oplus u\rangle = \left(\nabla^2\hat{\Psi} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\hat{\Psi}\right)|u^* \oplus u\rangle = 0 \tag{4.182}$$

and

$$\Box\hat{\Psi}^\star|u^* \oplus u\rangle = \left(\nabla^2\hat{\Psi} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\hat{\Psi}^\star\right)|u^* \oplus u\rangle = 0 \tag{4.183}$$

Finally we will investigate the entropy of the system. The Shannon entropy of a system is given by [21, 22],

$$H = -\sum_i P_i \log P_i\,, \tag{4.184}$$

where $P_i$ is the probability of the $i$-th state. The entropy for the both systems $|\Psi\rangle$ and $|\Psi^\star\rangle$ are

$$H(\theta) = -2\cos^2\theta\log|\cos\theta| - 2\sin^2\theta\log|\sin\theta|\,. \tag{4.185}$$

When the probability is equal the entropy is maximized, thus the $|\Psi(\pi/4)$ and $|\Psi^\star|(\pi/4)\rangle$ state is the equilibrium state, which is the most stable state. The system has 0.693 (corr. to 3 sig. fig.) bits. When it is either at its designated state with probability equal to 1, the entropy is zero, which is minimum and hence unstable. Therefore the Buddha is in His constant equilibrium state.

### 4.5.2 Canonical Quantization

In this section, we will dive into the core part of the whole subject, the quantization of duality wave, which is the quantization of thought (mind). Upon quantization, this will give the quantum of thought (mind). The procedure is called second quantization, which is originated from quantum field theory. Second quantization turns a wave into a particle quanta. Since our mind is created and annihilated periodically, it is described by the duality wave by the last section. Its quantization turn the mind into particle, this entails that which the external world is formed by our mind. This is to prove the fundamental theorem of Buddhism, every dharma is the realization of our thoughts.

First for the $\hat{\Psi}(\theta)$ we define the conjugated momentum operator by

$$\Pi = \frac{h}{i}\frac{\partial}{\partial\theta}\Psi\,. \tag{4.186}$$

Therefore the conjugated momentum of $\Psi$ is

$$\Pi(\theta) = \frac{h}{i}\Big(-\sin\theta(\hat{-}\oplus\hat{-}) + \cos\theta(\hat{+}\oplus\hat{+})\Big)\,. \tag{4.187}$$

Now we compute the commutator $[\hat{\Psi}(\theta), \hat{\Pi}(\theta)]$,

$$
\begin{aligned}
[\Psi(\theta), \Pi(\theta)] &= (-ih)[\cos\theta(\hat{-} \oplus \hat{-}) + \sin\theta(\hat{+} \oplus \hat{+})\ ,\ -\sin\theta(\hat{-} \oplus \hat{-}) + \cos\theta(\hat{+} \oplus \hat{+})] \\
&= -ih\Big( - \cos\theta \sin\theta[(\hat{-} \oplus \hat{-}), (\hat{-} \oplus \hat{-})] - \sin^2\theta[(\hat{+} \oplus \hat{+}), (\hat{-} \oplus \hat{-})] \\
&\quad + \cos^2\theta[(\hat{-} \oplus \hat{-}), (\hat{+} \oplus \hat{+})] + \sin\theta\cos\theta[(\hat{+} \oplus \hat{+}), (\hat{+} \oplus \hat{+})]\Big) \\
&= -ih[(\hat{-} \oplus \hat{-}), (\hat{+} \oplus \hat{+})] \\
&= -ih\Big((\hat{-} \oplus \hat{-})(\hat{+} \oplus \hat{+}) - (\hat{+} \oplus \hat{+})(\hat{-} \oplus \hat{-})\Big) \\
&= -ih(\hat{-}\hat{+} \oplus \hat{-}\hat{+} - \hat{+}\hat{-} \oplus \hat{+}\hat{-})
\end{aligned} \tag{4.188}
$$

Now we take an ansatz for the replacement of $\hat{-}, \hat{+}$ matrix operators by bosonic creation and annihilation operators $\hat{a}, \hat{a}^\dagger$ with commutation relation of $[\hat{a}, \hat{a}^\dagger] = 1$

$$\hat{-} \to \hat{a}^\dagger \quad , \quad \hat{+} \to \hat{a} \tag{4.189}$$

such that the quantum fields are

$$\hat{\Psi}(\theta) = \cos\theta(\hat{a}^\dagger \oplus \hat{a}^\dagger) + \sin\theta(\hat{a} \oplus \hat{a}) \tag{4.190}$$

and

$$\hat{\Pi}(\theta) = \frac{h}{i}\Big( - \sin\theta(\hat{a}^\dagger \oplus \hat{a}^\dagger) + \cos\theta(\hat{a} \oplus \hat{a})\Big)\,. \tag{4.191}$$

Then

$$
\begin{aligned}
\Big[\Psi\hat{(}\theta), \Pi\hat{(}\theta)\Big] &= -ih(\hat{a}^\dagger\hat{a} \oplus \hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger \oplus \hat{a}\hat{a}^\dagger) \\
&= -ih\Bigg( \begin{pmatrix} \hat{a}^\dagger\hat{a} & 0 \\ 0 & \hat{a}^\dagger\hat{a} \end{pmatrix} - \begin{pmatrix} \hat{a}\hat{a}^\dagger & 0 \\ 0 & \hat{a}\hat{a}^\dagger \end{pmatrix} \Bigg) \\
&= -ih \begin{pmatrix} -[\hat{a}, \hat{a}^\dagger] & 0 \\ 0 & -[\hat{a}, \hat{a}^\dagger] \end{pmatrix} \\
&= -ih \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \\
&= ih\mathbf{1}\,.
\end{aligned} \tag{4.192}
$$

In the meantime, we have $\Big[\hat{\Psi}(\theta), \hat{\Psi}(\theta)\Big] = 0$ and $\Big[\hat{\Pi}(\theta), \hat{\Pi}(\theta)\Big]$. Therefore we have the canonical quantization condition as

$$\Big[\hat{\Psi}(\theta), \hat{\Pi}(\theta)\Big] = ih\mathbf{1} \quad , \quad \Big[\hat{\Psi}(\theta), \hat{\Psi}(\theta)\Big] = \Big[\hat{\Pi}(\theta), \hat{\Pi}(\theta)\Big] = 0\,. \tag{4.193}$$

Next for the $\hat{\Phi}^\star(\theta)$ we define the conjugated momentum operator by

$$\hat{\Pi}^\star(\theta) = \frac{h}{i}\frac{\partial}{\partial\theta}\hat{\Psi}^\star\,. \tag{4.194}$$

Therefore the conjugated momentum of $\Psi$ is

$$\Pi^\star(\theta) = \frac{h}{i}\Big( - \sin\theta(\hat{+} \oplus \hat{-}) + \cos\theta(\hat{-} \oplus \hat{+})\Big)\,. \tag{4.195}$$

Now we compute the commutator $[\hat{\Psi}(\theta), \hat{\Pi}(\theta)]$,

$$\begin{aligned}
[\Psi^\star(\theta), \Pi^\star(\theta)] &= (-ih)\left[\cos\theta(\hat{+}\oplus\hat{-}) + \sin\theta(\hat{-}\oplus\hat{+})\ ,\ -\sin\theta(\hat{+}\oplus\hat{-}) + \cos\theta(\hat{-}\oplus\hat{+})\right] \\
&= -ih\Big(-\cos\theta\sin\theta\left[(\hat{+}\oplus\hat{-}), (\hat{+}\oplus\hat{-})\right] - \sin^2\theta\left[(\hat{-}\oplus\hat{+}), (\hat{+}\oplus\hat{-})\right] \\
&\quad + \cos^2\theta\left[(\hat{+}\oplus\hat{-}), (\hat{-}\oplus\hat{+})\right] + \sin\theta\cos\theta\left[(\hat{-}\oplus\hat{+}), (\hat{-}\oplus\hat{+})\right]\Big) \\
&= -ih\left[(\hat{+}\oplus\hat{-}), (\hat{-}\oplus\hat{+})\right] \\
&= -ih\Big((\hat{+}\oplus\hat{-})(\hat{-}\oplus\hat{+}) - (\hat{-}\oplus\hat{+})(\hat{+}\oplus\hat{-})\Big) \\
&= -ih(\hat{+}\hat{-}\oplus\hat{-}\hat{+} - \hat{-}\hat{+}\oplus\hat{+}\hat{-})
\end{aligned} \tag{4.196}$$

Then we have

$$\begin{aligned}
\left[\hat{\Psi}^\star(\theta), \hat{\Pi}^\star(\theta)\right] &= -ih(\hat{a}\hat{a}^\dagger \oplus \hat{a}^\dagger\hat{a} - \hat{a}^\dagger\hat{a} \oplus \hat{a}\hat{a}^\dagger) \\
&= -ih\left(\begin{pmatrix}\hat{a}\hat{a}^\dagger & 0 \\ 0 & \hat{a}^\dagger\hat{a}\end{pmatrix} - \begin{pmatrix}\hat{a}^\dagger\hat{a} & 0 \\ 0 & \hat{a}\hat{a}^\dagger\end{pmatrix}\right) \\
&= -ih\begin{pmatrix}[\hat{a}, \hat{a}^\dagger] & 0 \\ 0 & -[\hat{a}, \hat{a}^\dagger]\end{pmatrix} \\
&= ih\begin{pmatrix}-1 & 0 \\ 0 & 1\end{pmatrix} \\
&= ihM\,,
\end{aligned} \tag{4.197}$$

where $M$ is another representation matrix of the duality operator with $M^2 = \mathbf{1}$. In the meantime, we have $[\hat{\Psi}^\star(\theta), \hat{\Psi}^\star(\theta)] = 0$ and $[\hat{\Pi}^\star(\theta), \hat{\Pi}^\star(\theta)] = 0$. Therefore we have the canonical quantization condition as

$$[\hat{\Psi}^\star(\theta), \hat{\Pi}^\star(\theta)] = ihM \quad , \quad [\hat{\Psi}^\star(\theta), \hat{\Psi}^\star(\theta)] = [\hat{\Pi}^\star(\theta), \hat{\Pi}^\star(\theta)] = 0\,. \tag{4.198}$$

Notice that the cross terms are not zero, this is given by the fact that the states are not independent of each other. For example

$$[\hat{\Psi}(\theta), \hat{\Psi}^\star(\theta)] = \begin{pmatrix}-\cos 2\theta & 0 \\ 0 & 0\end{pmatrix}\,. \tag{4.199}$$

**Hamiltonian**

The canonical momentum operator was defined by $\hat{\Pi} = \frac{1}{i}\frac{\partial}{\partial\theta}$, the $h$ is introduced such that we will get the same form of commutation relation as quantum mechanics does. From now on, to calculate the canonical Hamiltonian, let's define the dimensionless conjugated momentum, still denoted as $\hat{\Pi}$,

$$\hat{\Pi} = \frac{1}{i}\frac{\partial}{\partial\theta}\hat{\Psi} \quad \text{and} \quad \hat{\Pi}^\star = \frac{1}{i}\frac{\partial}{\partial\theta}\hat{\Psi}^\star \tag{4.200}$$

We ansatz that the Hamiltonian is defined as

$$\hat{H} = \left(\frac{1}{2}\hat{\Pi}^2 + \frac{1}{2}\hat{\Psi}^2\right)\hbar\omega \quad \text{and} \quad \hat{H}^\star = \left(\frac{1}{2}\hat{\Pi}^{\star 2} + \frac{1}{2}\hat{\Psi}^{\star 2}\right)\hbar\omega\,. \tag{4.201}$$

Now first we evaluate $\hat{H}(\theta)$.

$$
\begin{aligned}
&\hat{\Pi}^2(\theta)\\
&= \frac{1}{2i^2}\Big( -\sin\theta(\hat{-}\oplus\hat{-}) + \cos\theta(\hat{+}\oplus\hat{+})\Big)\Big( -\sin\theta(\hat{-}\oplus\hat{-}) + \cos\theta(\hat{+}\oplus\hat{+})\Big)\\
&= -\frac{1}{2}\Big( \sin^2\theta(\hat{-}\oplus\hat{-})(\hat{-}\oplus\hat{-}) - \sin\theta\cos\theta(\hat{-}\oplus\hat{-})(\hat{+}\oplus\hat{+})\\
&\qquad - \cos\theta\sin\theta(\hat{+}\oplus\hat{+})(\hat{-}\oplus\hat{-}) + \cos^2\theta(\hat{+}\oplus\hat{+})(\hat{+}\oplus\hat{+})\Big)\\
&= -\frac{1}{2}\Big( \sin^2\theta(\hat{-}\hat{-}\oplus\hat{-}\hat{-}) - \sin\theta\cos\theta(\hat{-}\hat{+}\oplus\hat{-}\hat{+}) - \sin\theta\cos\theta(\hat{+}\hat{-}\oplus\hat{+}\hat{-}) + \cos^2\theta(\hat{+}\hat{+}\oplus\hat{+}\hat{+})\Big)\\
&= -\frac{1}{2}\Bigg[(\sin^2\theta\begin{pmatrix}\hat{-}\hat{-} & 0\\ 0 & \hat{-}\hat{-}\end{pmatrix} - \sin\theta\cos\theta\begin{pmatrix}\hat{-}\hat{+} & 0\\ 0 & \hat{-}\hat{+}\end{pmatrix} - \sin\theta\cos\theta\begin{pmatrix}\hat{+}\hat{-} & 0\\ 0 & \hat{+}\hat{-}\end{pmatrix} + \cos^2\theta\begin{pmatrix}\hat{+}\hat{+} & 0\\ 0 & \hat{+}\hat{+}\end{pmatrix}\Bigg]\\
&= -\frac{1}{2}\begin{pmatrix}\sin^2\theta\hat{-}\hat{-} + \cos^2\theta\hat{+}\hat{+} - \sin\theta\cos\theta(\hat{-}\hat{+} + \hat{+}\hat{-}) & 0\\ 0 & \sin^2\theta\hat{-}\hat{-} + \cos^2\theta\hat{+}\hat{+} - \sin\theta\cos\theta(\hat{-}\hat{+} + \hat{+}\hat{-})\end{pmatrix}\\
&= -\frac{1}{2}\Big( \sin^2\theta\hat{-}\hat{-} + \cos^2\theta\hat{+}\hat{+} - \sin\theta\cos\theta(\hat{-}\hat{+} + \hat{+}\hat{-})\Big)\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\\
&= -\frac{1}{2}\Big( \sin^2\theta\hat{a}^\dagger\hat{a}^\dagger + \cos^2\theta\hat{a}\hat{a} - \sin\theta\cos\theta(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})\Big)\mathbf{1}\\
&= -\frac{1}{2}\Big( \sin^2\theta\hat{a}^\dagger\hat{a}^\dagger + \cos^2\theta\hat{a}\hat{a}\Big) + \frac{1}{2}(2\sin\theta\cos\theta)\Big(\hat{a}^\dagger\hat{a} + \frac{1}{2}\Big)\mathbf{1}\\
&= -\frac{1}{2}\Big( \sin^2\theta\hat{a}^\dagger\hat{a}^\dagger + \cos^2\theta\hat{a}\hat{a}\Big) + \sin\theta\cos\theta\Big(\hat{a}^\dagger\hat{a} + \frac{1}{2}\Big)\mathbf{1}\,.
\end{aligned}
\tag{4.202}
$$

Then

$$
\begin{aligned}
&\hat{\Psi}^2(\theta)\\
&= \frac{1}{2}\Big( \cos\theta(\hat{-}\oplus\hat{-}) + \sin\theta(\hat{+}\oplus\hat{+})\Big)\Big( \cos\theta(\hat{-}\oplus\hat{-}) + \sin\theta(\hat{+}\oplus\hat{+})\Big)\\
&= \frac{1}{2}\Big( \cos^2\theta(\hat{-}\oplus\hat{-})(\hat{-}\oplus\hat{-}) + \cos\theta\sin\theta(\hat{-}\oplus\hat{-})(\hat{+}\oplus\hat{+})\\
&\qquad + \sin\theta\cos\theta(\hat{+}\oplus\hat{+})(\hat{-}\oplus\hat{-}) + \sin^2\theta(\hat{+}\oplus\hat{+})(\hat{+}\oplus\hat{+})\Big)\\
&= \frac{1}{2}\Big( \cos^2\theta(\hat{-}\hat{-}\oplus\hat{-}\hat{-}) + \cos\theta\sin\theta(\hat{-}\hat{+}\oplus\hat{-}\hat{+}) + \cos\theta\sin\theta(\hat{+}\hat{-}\oplus\hat{+}\hat{-}) + \sin^2\theta(\hat{+}\hat{+}\oplus\hat{+}\hat{+})\Big)\\
&= \frac{1}{2}\Bigg[(\cos^2\theta\begin{pmatrix}\hat{-}\hat{-} & 0\\ 0 & \hat{-}\hat{-}\end{pmatrix} + \cos\theta\sin\theta\begin{pmatrix}\hat{-}\hat{+} & 0\\ 0 & \hat{-}\hat{+}\end{pmatrix} + \cos\theta\sin\theta\begin{pmatrix}\hat{+}\hat{-} & 0\\ 0 & \hat{+}\hat{-}\end{pmatrix} + \sin^2\theta\begin{pmatrix}\hat{+}\hat{+} & 0\\ 0 & \hat{+}\hat{+}\end{pmatrix}\Bigg]\\
&= \frac{1}{2}\begin{pmatrix}\cos^2\theta\hat{-}\hat{-} + \sin^2\theta\hat{+}\hat{+} + \cos\theta\sin\theta(\hat{-}\hat{+} + \hat{+}\hat{-}) & 0\\ 0 & \cos^2\theta\hat{-}\hat{-} + \sin^2\theta\hat{+}\hat{+} + \cos\theta\sin\theta(\hat{-}\hat{+} + \hat{+}\hat{-})\end{pmatrix}\\
&= \frac{1}{2}\Big( \cos^2\theta\hat{-}\hat{-} + \sin^2\theta\hat{+}\hat{+} + \cos\theta\sin\theta(\hat{-}\hat{+} + \hat{+}\hat{-})\Big)\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\\
&= \frac{1}{2}\Big( \cos^2\theta\hat{a}^\dagger\hat{a}^\dagger + \sin^2\theta\hat{a}\hat{a} + \cos\theta\sin\theta(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})\Big)\mathbf{1}\\
&= \frac{1}{2}\Big( \cos^2\theta\hat{a}^\dagger\hat{a}^\dagger + \sin^2\theta\hat{a}\hat{a}\Big)\mathbf{1} + \frac{1}{2}(2\cos\theta\sin\theta)\Big(\hat{a}^\dagger\hat{a} + \frac{1}{2}\Big)\mathbf{1}\\
&= \frac{1}{2}\Big( \cos^2\theta\hat{a}^\dagger\hat{a}^\dagger + \sin^2\theta\hat{a}\hat{a}\Big)\mathbf{1} + \sin\theta\cos\theta\Big(\hat{a}^\dagger\hat{a} + \frac{1}{2}\Big)\mathbf{1}\,.
\end{aligned}
\tag{4.203}
$$

Therefore, finally we obtain the Hamiltonian as follow

$$\hat{H}(\theta) = \left(\frac{1}{2}\hat{\Pi}^2(\theta) + \frac{1}{2}\hat{\Psi}^2(\theta)\right)\hbar\omega = \frac{1}{2}\cos 2\theta(\hat{a}^\dagger\hat{a}^\dagger - \hat{a}\hat{a})\hbar\omega\mathbf{1} + \sin 2\theta\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\hbar\omega\mathbf{1} \tag{4.204}$$

The expectation energy is hence given by

$$\begin{aligned}\langle E\rangle = \langle n|\hat{H}(\theta)|n\rangle &= \frac{1}{2}\cos 2\theta\langle n|(\hat{a}^\dagger\hat{a}^\dagger - \hat{a}\hat{a})|n\rangle\hbar\omega\,\mathbf{1} + \sin 2\theta\langle n|\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)|n\rangle\hbar\omega\mathbf{1} \\ &= \sin 2\theta\left(n + \frac{1}{2}\right)\hbar\omega\mathbf{1} \\ &= H\sin 2\theta\,\mathbf{1}\,,\end{aligned} \tag{4.205}$$

where

$$H = \left(n + \frac{1}{2}\right)\hbar\omega. \tag{4.206}$$

This is because $\langle n|\hat{a}^\dagger\hat{a}^\dagger|n\rangle = \langle n|\hat{a}\hat{a}|n\rangle = 0$ and $\langle n|\hat{a}^\dagger\hat{a}|n\rangle = n$. The vacuum expectation energy is given by

$$\langle 0|\hat{H}(\theta)|0\rangle = \frac{1}{2}\sin 2\theta\hbar\omega\,\mathbf{1} = \sin\theta\cos\theta\hbar\omega\,\mathbf{1}\,. \tag{4.207}$$

Now let's evaluate the vacuum expectation value for different important phases. When $\theta = 0$ or $\frac{\pi}{2}$, we have

$$E_0 = \langle 0|\hat{H}(\theta)|0\rangle = \mathbf{0}\,. \tag{4.208}$$

Recalling that $\Psi(\theta) = \cos\theta|0\oplus 0\rangle + \sin\theta|1\oplus 1\rangle = \cos\theta|\text{nothingness}\rangle + \sin\theta|\text{ālaya-vijñāna}\rangle$ , this means when the dual wave function is at a particular state, then it has zero vacuum expectation energy value. On the other hand, when $\theta = \pi/4$,

$$E_0 = \langle 0|\hat{H}(\theta)|0\rangle = \frac{1}{2}\hbar\omega\,\mathbf{1} = \begin{pmatrix}\frac{1}{2}\hbar\omega & 0 \\ 0 & \frac{1}{2}\hbar\omega\end{pmatrix}\,. \tag{4.209}$$

This means that when $\Psi(\frac{\pi}{4}) = \frac{1}{\sqrt{2}}|0\oplus 0\rangle + \frac{1}{\sqrt{2}}|1\oplus 1\rangle = \frac{1}{\sqrt{2}}|\text{nothingness}\rangle + \frac{1}{\sqrt{2}}|\text{ālaya-vijñāna}\rangle$, the state is in linear superposition of the two states has its greatest entropy, it has the maximized energy. This shows that when the system is in its determined state of $|0\oplus 0\rangle$ or $|1\oplus 1\rangle$ it has lower ground state energy than the undetermined superpositioned state of $\Psi = \frac{1}{\sqrt{2}}|0\oplus 0\rangle + \frac{1}{\sqrt{2}}|1\oplus 1\rangle$, thus is considered to be more stable. This implies that either the $|\text{sentient being}\rangle$ state or the $|\text{Buddha}\rangle$ state is more stable. We say the superpositioned state is spontaneously collapsed into one of the state of full probability, this statement is equivalent to spontaneous symmetry breaking when a state is completely determined.

Next we evaluate $\hat{H}^{\star}(\theta)$.

$$
\begin{aligned}
&\hat{\Pi}^{\star 2}(\theta) \\
&= \frac{1}{2i^2}\Big(-\sin\theta(\hat{+}\oplus\hat{-})+\cos\theta(\hat{-}\oplus\hat{+})\Big)\Big(-\sin\theta(\hat{+}\oplus\hat{-})+\cos\theta(\hat{-}\oplus\hat{+})\Big) \\
&= -\frac{1}{2}\Big(\sin^2\theta(\hat{+}\oplus\hat{-})(\hat{+}\oplus\hat{-})-\sin\theta\cos\theta(\hat{+}\oplus\hat{-})(\hat{-}\oplus\hat{+}) \\
&\qquad -\cos\theta\sin\theta(\hat{-}\oplus\hat{+})(\hat{+}\oplus\hat{-})+\cos^2\theta(\hat{-}\oplus\hat{+})(\hat{-}\oplus\hat{+})\Big) \\
&= -\frac{1}{2}\Big(\sin^2\theta(\hat{+}\hat{+}\oplus\hat{-}\hat{-})-\sin\theta\cos\theta(\hat{+}\hat{-}\oplus\hat{-}\hat{+})-\sin\theta\cos\theta(\hat{-}\hat{+}\oplus\hat{+}\hat{-})+\cos^2\theta(\hat{-}\hat{-}\oplus\hat{+}\hat{+})\Big) \\
&= -\frac{1}{2}\left[(\sin^2\theta\begin{pmatrix}\hat{+}\hat{+} & 0\\ 0 & \hat{-}\hat{-}\end{pmatrix}-\sin\theta\cos\theta\begin{pmatrix}\hat{+}\hat{-} & 0\\ 0 & \hat{-}\hat{+}\end{pmatrix}-\sin\theta\cos\theta\begin{pmatrix}\hat{-}\hat{+} & 0\\ 0 & \hat{+}\hat{-}\end{pmatrix}+\cos^2\theta\begin{pmatrix}\hat{-}\hat{-} & 0\\ 0 & \hat{+}\hat{+}\end{pmatrix}\right] \\
&= -\frac{1}{2}\begin{pmatrix}\sin^2\theta\hat{+}\hat{+}+\cos^2\theta\hat{-}\hat{-}-\sin\theta\cos\theta(\hat{+}\hat{-}+\hat{-}\hat{+}) & 0\\ 0 & \sin^2\theta\hat{-}\hat{-}+\cos^2\theta\hat{+}\hat{+}-\sin\theta\cos\theta(\hat{-}\hat{+}+\hat{+}\hat{-})\end{pmatrix} \\
&= -\frac{1}{2}\begin{pmatrix}\sin^2\theta\hat{a}\hat{a}+\cos^2\theta\hat{a}^\dagger\hat{a}^\dagger-\sin\theta\cos\theta(\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}) & 0\\ 0 & \sin^2\theta\hat{a}^\dagger\hat{a}^\dagger+\cos^2\theta\hat{a}\hat{a}-\sin\theta\cos\theta(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)\end{pmatrix} \\
&= -\frac{1}{2}\begin{pmatrix}\sin^2\theta\hat{a}\hat{a}+\cos^2\theta\hat{a}^\dagger\hat{a}^\dagger-\sin 2\theta\Big(\hat{a}^\dagger\hat{a}+\frac{1}{2}\Big) & 0\\ 0 & \sin^2\theta\hat{a}^\dagger\hat{a}^\dagger+\cos^2\theta\hat{a}\hat{a}-\sin 2\theta\Big(\hat{a}^\dagger\hat{a}+\frac{1}{2}\Big)\end{pmatrix}.
\end{aligned}
\tag{4.210}
$$

Then

$$
\begin{aligned}
&\hat{\Psi}^{\star 2}(\theta) \\
&= \frac{1}{2}\Big(\cos\theta(\hat{+}\oplus\hat{-})+\sin\theta(\hat{-}\oplus\hat{+})\Big)\Big(\cos\theta(\hat{+}\oplus\hat{-})+\sin\theta(\hat{-}\oplus\hat{+})\Big) \\
&= \frac{1}{2}\Big(\cos^2\theta(\hat{+}\oplus\hat{-})(\hat{+}\oplus\hat{-})+\cos\theta\sin\theta(\hat{+}\oplus\hat{-})(\hat{-}\oplus\hat{+}) \\
&\qquad +\sin\theta\cos\theta(\hat{-}\oplus\hat{+})(\hat{+}\oplus\hat{-})+\sin^2\theta(\hat{-}\oplus\hat{+})(\hat{-}\oplus\hat{+})\Big) \\
&= \frac{1}{2}\Big(\cos^2\theta(\hat{+}\hat{+}\oplus\hat{-}\hat{-})+\cos\theta\sin\theta(\hat{+}\hat{-}\oplus\hat{-}\hat{+})+\cos\theta\sin\theta(\hat{-}\hat{+}\oplus\hat{+}\hat{-})+\sin^2\theta(\hat{-}\hat{-}\oplus\hat{+}\hat{+})\Big) \\
&= \frac{1}{2}\left[(\cos^2\theta\begin{pmatrix}\hat{+}\hat{+} & 0\\ 0 & \hat{-}\hat{-}\end{pmatrix}+\cos\theta\sin\theta\begin{pmatrix}\hat{+}\hat{-} & 0\\ 0 & \hat{-}\hat{+}\end{pmatrix}+\cos\theta\sin\theta\begin{pmatrix}\hat{-}\hat{+} & 0\\ 0 & \hat{+}\hat{-}\end{pmatrix}+\sin^2\theta\begin{pmatrix}\hat{-}\hat{-} & 0\\ 0 & \hat{+}\hat{+}\end{pmatrix}\right] \\
&= \frac{1}{2}\begin{pmatrix}\cos^2\theta\hat{+}\hat{+}+\sin^2\theta\hat{-}\hat{-}+\cos\theta\sin\theta(\hat{+}\hat{-}+\hat{-}\hat{+}) & 0\\ 0 & \cos^2\theta\hat{-}\hat{-}+\sin^2\theta\hat{+}\hat{+}+\cos\theta\sin\theta(\hat{-}\hat{+}+\hat{+}\hat{-})\end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix}\cos^2\theta\hat{a}\hat{a}+\sin^2\theta\hat{a}^\dagger\hat{a}^\dagger+\cos\theta\sin\theta(\hat{a}\hat{a}^\dagger+\hat{a}^\dagger\hat{a}) & 0\\ 0 & \cos^2\theta\hat{a}^\dagger\hat{a}^\dagger+\sin^2\theta\hat{a}\hat{a}+\cos\theta\sin\theta(\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger)\end{pmatrix} \\
&= \frac{1}{2}\begin{pmatrix}\cos^2\theta\hat{a}\hat{a}+\sin^2\theta\hat{a}^\dagger\hat{a}^\dagger+\sin 2\theta\Big(\hat{a}^\dagger\hat{a}+\frac{1}{2}\Big) & 0\\ 0 & \cos^2\theta\hat{a}^\dagger\hat{a}^\dagger+\sin^2\theta\hat{a}\hat{a}+\sin 2\theta\Big(\hat{a}^\dagger\hat{a}+\frac{1}{2}\Big)\end{pmatrix}.
\end{aligned}
\tag{4.211}
$$

Therefore, finally we obtain the Hamiltonian as follow

$$
\begin{aligned}
&\hat{H}^{\star}(\theta) = \left(\frac{1}{2}\hat{\Pi}^{\star 2}(\theta) + \frac{1}{2}\hat{\Psi}^{\star 2}(\theta)\right)\hbar\omega \\
&= \begin{pmatrix} \frac{1}{2}\cos 2\theta(\hat{a}\hat{a} - \hat{a}^{\dagger}\hat{a}^{\dagger}) + \sin 2\theta\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right) & 0 \\ 0 & \frac{1}{2}\cos 2\theta(\hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}\hat{a}) + \sin 2\theta\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right) \end{pmatrix}\hbar\omega \\
&= \frac{1}{2}\cos 2\theta(\hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}\hat{a})\hbar\omega M + \sin 2\theta\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right)\hbar\omega\mathbf{1}\,.
\end{aligned}
\tag{4.212}
$$

The expectation energy is hence given by

$$
\begin{aligned}
\langle E^{\star}\rangle = \langle n|\hat{H}^{\star}(\theta)|n\rangle &= \frac{1}{2}\cos 2\theta\langle n|(\hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}\hat{a})|n\rangle h\omega M + \sin 2\theta\langle n|\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right)|n\rangle h\omega\mathbf{1} \\
&= \sin 2\theta\left(n + \frac{1}{2}\right)\hbar\omega\mathbf{1} \\
&= H\sin 2\theta\,\mathbf{1}\,,
\end{aligned}
\tag{4.213}
$$

Therefore the expectation energy of $\Psi^{*}(\theta)$ is same as the case of $\Psi(\theta)$. That means

$$
\langle E(\theta)\rangle = \langle E^{\star}(\theta)\rangle = H\sin 2\theta\mathbf{1}\,. \tag{4.214}
$$

Hence the phase analysis remains the same as the $\Psi(\theta)$ case.

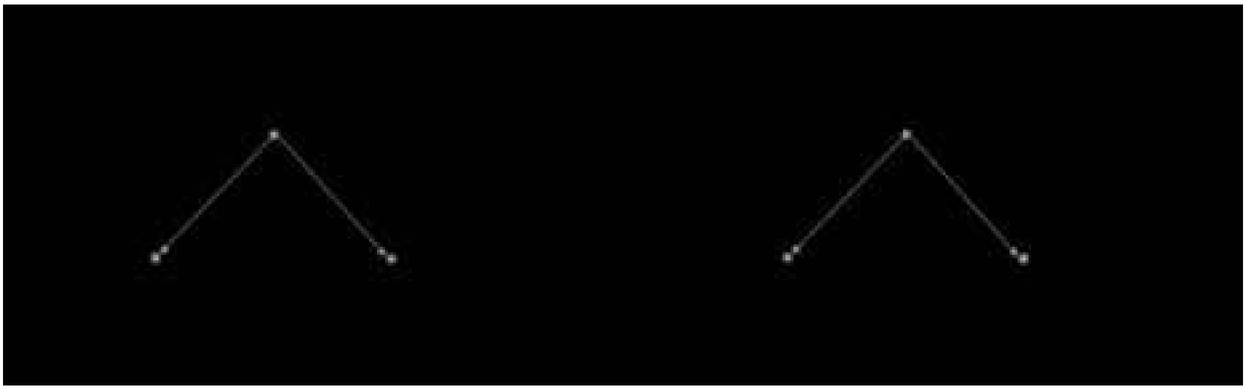

Figure 4.18: For the left figure, we compute the ground state n = 0 for phase = =4; 0; =2, corresponding to $|\Psi(\ =4)\rangle = \frac{1}{\sqrt{2}}(|0\oplus 0\rangle + |1\oplus 1\rangle)$; $|\Psi(0)\rangle = |0\oplus 0\rangle$ and $|\Psi(\ =2)\rangle = |1\oplus 1\rangle$ respectively. The $|\Psi(\ =4)\rangle$ is a quantum state with probability of each state of 1=2, it has the average ground state energy of $\frac{1}{2}\hbar!$ , while $|\Psi(0)\rangle$ and $|\Psi(\ =2)\rangle$ are classical states, with probability equal to 1 with average ground energy equal to zero. The idea is same for the right figure.

In the Hamiltonian, we have the extra term of $\hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}\hat{a}$ in addition to the standard Hamiltonian of simple harmonic oscillator. We would like to investigate its meaning here. First consider

$$
\begin{aligned}
&(\hat{a}^{\dagger} + \hat{a})(\hat{a}^{\dagger} - \hat{a}) \\
&= \hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}^{\dagger}\hat{a} + \hat{a}\hat{a}^{\dagger} - \hat{a}\hat{a} \\
&= \hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}\hat{a} + [\hat{a}, \hat{a}^{\dagger}] \\
&= \hat{a}^{\dagger}\hat{a}^{\dagger} - \hat{a}\hat{a} + 1
\end{aligned}
\tag{4.215}
$$